

# Jewels: Toward Text-Conditioned, Streamable, and Locally Editable Video as Space-Time Gaussian Fields

The Jewels Project

Technical report – August 14, 2026

**Status.** This is a progress report, not a benchmark or a claim of completed text-to-video generation. It separates established results, failed controls, and hypotheses still under test.

## Abstract

We investigate representing ordinary RGB video as an explicit field of anisotropic Gaussian primitives (*jewels*) in normalized image-time coordinates: each primitive occupies a volume in (u,v,t), carries opacity and a linear color field, and tilts through time so that motion is oriented space-time support rather than frame-to-frame change. The goal is a promptable text-to-video model whose generation is the *composition* of such primitives — persistent across windows by stable identity, editable by direct manipulation, rendered deterministically — with semantics supplied during training by a frozen pretrained video prior (LTX-2.3) through a coarse scaffold raster of about six kilobytes per 16-frame stride.

Four results define the current state. First, the representation itself is not the bottleneck: optimized 72k-jewel fields replay held-out generated videos at 28–35 dB and 0.098 LPIPS — near-photoreal perceptual quality. Second, direct set-space generation was measured to a wall and replaced: every mark-space sampler — deterministic, stochastic, and two coupling variants — rendered worse macro-structure than a no-model trilinear decode of its own scaffold input, across every corpus size and domain, a joint-composition failure our metric triangulation (reference, perceptual, and texture-blind layout) localizes precisely. The selected architecture is instead a dense intermediate: a ~5M feed-forward encoder performs amortized fitting of teacher-generated video into jewel fields and one-shots held-out clips at 23.4 dB / 0.944 SSIM / 0.405 LPIPS with macro-layout at the fitted ceiling — the first arm to beat that trilinear baseline on every metric, eight decibels beyond the best set-generative arm, from twelve training windows. Third, quality scales with data and is priced in compute: a frozen-protocol curve over 4/8/12 fitted training sources is monotone in the mean, with a measured three-seed noise floor resolving its first step at three standard deviations; training on teacher-generated fits removes a cross-domain feature-loss tax of one third at thirteen standard deviations; and a mixed real-plus-teacher corpus shows the two supervision sources are complementary (texture versus macro-structure). One fitted source costs about 25 GPU-minutes and the teacher supplies prompt-matched sources without limit. Fourth, direct text selectivity remains open: correct and shuffled prompts tie once the video scaffold is present, so prompt semantics currently live in the teacher, and distilling the teacher’s coarse scaffold into a compact text-conditioned generator is the identified route to prompt-only inference.

Gaussian reconstruction of video is established art (VeGaS, GSVF, GaussianVideo, GaRV); the narrower opportunity evaluated here is a corpus-level generative distribution over persistent, editable space-time Gaussian fields. This is a progress report: negative results, falsified hypotheses, and failed visual audits are reported as first-class evidence, and no claim is made that current samples are competitive with raster video generation.

## 1 Introduction

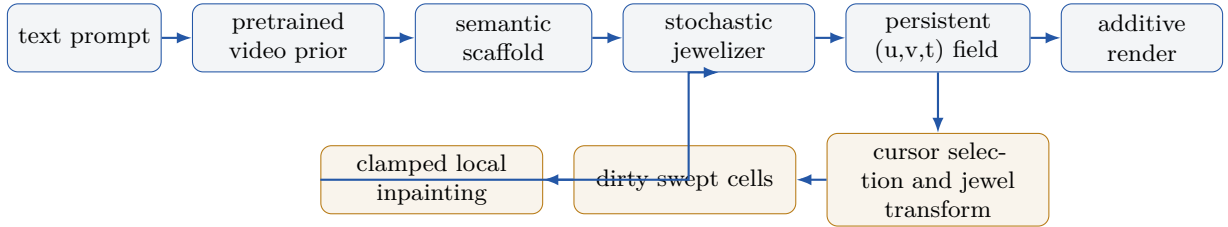
Pixel video models repeatedly synthesize dense frames and must learn temporal coherence through architecture, loss, or data. JEWELS asks whether coherence can instead be partly structural. We treat a video as a continuous three-dimensional signal over horizontal position, vertical position, and time. A moving visual element can then be represented by a Gaussian tube whose principal axis leans through time. The model is explicit: primitives can be selected, moved, recolored, carried between generation windows, culled at multiple levels of detail, and rendered at arbitrary times.

The intended system is not merely a codec. It is a text-to-video model whose native output is an editable space-time field. After generation, a viewer sees the video as a parallelepiped, drags a cursor through it, moves one or many jewels, and asks the model to resample only the vacated and destination neighborhoods. Untouched regions are hard constraints rather than soft conditioning.

The project has progressed through a sequence of falsifiable gates:

1. demonstrate that anisotropic space-time primitives can fit motion and ordinary video;
2. measure whether fitted sets are canonical enough for a shared generative model;
3. raise effective splat density without destroying temporal persistence;
4. learn a deterministic, count-aware local bottleneck for dense fields;
5. generate hierarchical latents with tractable attention;
6. prove exact edit locality and overlap-conditioned continuation; and
7. introduce text only after a leakage-safe prompt corpus exists.

Our contributions are therefore empirical and architectural rather than a completed generative benchmark. We report (i) an additive (u,v,t) Gaussian representation and gauge-free field features, (ii) density and tokenizer experiments that identify what preserves small moving subjects, (iii) controlled evidence that spatially aligned prefix context is essential for continuation, (iv) an edit/inpaint formulation with exact clean-region preservation, and (v) explicit negative results that bound what remains unproved. We further isolate deterministic conditional averaging as the current washout mechanism and show that a low-resolution semantic scaffold can recover global layout in stochastic jewel realization and transfer from UCF training to unseen prompt-generated LTX videos without mark-model retraining. We then replace fitted continuation topology with scaffold/carry-predicted counts and show that the frozen realizer remains within 0.028 dB of its oracle-topology control at the intended effective density. Finally, we remove fitted initial state, run three autonomous windows with exact append-only identity, isolate a lifecycle/appearance coupling that scalar foreground weighting does not resolve, and pass a conservative factorized control whose scaffold-gated RGB residual cannot alter lifecycle, topology, density, or identity. We then compress that path into a 74k-parameter base-locked adapter; exact ownership survives, but the student reproduces only part of the teacher’s fidelity gain. A frozen-base neighborhood set block subsequently produces broad fidelity/error gains under exact base-owned topology, while failing the stricter structural/visual gate and selecting rendered set/trajectory supervision as the next bounded test. We additionally articulate two falsifiable systems hypotheses: that explicit persistent fields offer interaction properties unavailable from an opaque raster latent alone, and that a hierarchical field can serve as a multimodal visual event language.



**Figure 1:** Revised target system. A pretrained raster video prior supplies global semantics; the jewelizer emits the persistent explicit state. Editing moves protected primitives, marks source, destination, and swept cells dirty, and resamples only those cells while the rest remains exact.

## 2 Relationship to Prior Work and Claim Scope

Modern Gaussian splatting was popularized for explicit 3D radiance fields by Kerbl et al. [8]. Dynamic-view methods subsequently added deformation, time, or native higher-dimensional primitives [10, 27]. Those systems reconstruct a view-dependent 3D scene; our target is the displayed 2D RGB video itself, without recovering physical depth or novel views.

That narrower video-representation lane is also established. Splatter a Video embeds monocular video into canonical 3D Gaussians with motion [22]. VeGaS uses folded Gaussian distributions to obtain time-conditioned 2D Gaussians and supports editing [20]. GSVC predicts frame-level 2D Gaussian sets from previous frames for compression [24]. GaussianVideo learns time-varying deformations of 2D Gaussians [9]. Most directly, GaRV encodes videos as 3D Gaussians in a space-time volume [19]. HyperGS amortizes reconstruction by predicting per-frame Gaussian representations feed-forward from input video [31]. Therefore, **we do not claim to be first to reconstruct 2D video with Gaussian splats, nor first to use a Gaussian space-time volume.**

Generative Gaussian work is substantial but usually targets 3D objects, rooms, or dynamic 3D scenes. GaussianCube and L3DG learn structured or latent generative models of 3D Gaussians [18, 28]. VideoRFSplat generates text-

conditioned 3D scenes with a video model [4]; Diff4Splat predicts a deformable 3D Gaussian scene from an image, camera path, and optional text [15]. DiffSplat repurposes a large image diffusion prior to generate Gaussian splat grids and reports a rendering loss as important for coherence across novel views [11]. This supports inheriting mature raster priors and supervising rendered output rather than treating independent Gaussian-parameter losses as a complete appearance objective.

The jewel births are also a marked spatiotemporal point set. Point Set Diffusion directly models conditional spatial and spatiotemporal point processes without an intensity-function factorization [14], while point-cloud flow work emphasizes that permutation invariance and the coupling between noise and unordered targets materially affect the learned transport [7]. These results motivate a stochastic set objective; they do not remove the need for a coherent video-level condition in the present task.

Our provisional distinction is the combination of: (1) direct generation of the displayed 2D video as an anisotropic (u,v,t) Gaussian field, (2) persistent stable-ID carry and birth emission across overlapping time windows, and (3) hard-local jewel manipulation followed by constrained latent repair. Our August 2026 search found no published system with this exact combination. However, absence from a search is not proof of priority, and the current project has not yet shown generalizing text-to-video generation. We therefore use “toward” in the title and reserve any first-of-kind claim for a completed, externally benchmarked model and a broader review.

**Table 1:** Positioning against the closest video-Gaussian families. “Amortized” means a shared encoder predicts a representation from input video; it is not the same as unconditional or text-conditioned generation from noise.

| Method          | Native target | Temporal form                           | Shared prediction      | Primary goal           |
|-----------------|---------------|-----------------------------------------|------------------------|------------------------|
| VeGaS           | 2D video      | folded 3D Gaussians, conditional slices | no                     | representation/editing |
| GSVC            | 2D video      | linked per-frame 2D splats              | no                     | compression            |
| GaussianVideo   | 2D video      | time-deformed 2D splats                 | no                     | representation; codec  |
| GaRV            | 2D video      | 3D space-time Gaussians                 | encoder/decoder scheme | random-access decoding |
| HyperGS         | 2D video      | predicted per-frame 2D splats           | yes, from video        | amortized encoding     |
| JEWELS (target) | 2D video      | persistent anisotropic (u,v,t) field    | from noise and text    | generation/editing     |

## 3 Method

### 3.1 Video as an additive space-time field

Let the normalized video domain be  $\Omega = [-1, 1]^3$ , with a query  $\mathbf{x} = (u, v, t) \in \Omega$ . Jewel  $i$  has center  $\boldsymbol{\mu}_i \in \mathbb{R}^3$ , symmetric positive definite covariance  $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3 \times 3}$ , weight logit  $\ell_i$ , base color  $\mathbf{c}_i \in \mathbb{R}^3$ , and a first-order color matrix  $\mathbf{G}_i \in \mathbb{R}^{3 \times 3}$ . Its contribution is

$$q_i(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i), \quad (1)$$

$$a_i(\mathbf{x}) = \sigma(\ell_i) \exp[-q_i(\mathbf{x})/2], \quad (2)$$

$$\hat{\mathbf{I}}(\mathbf{x}) = \mathbf{b} + \sum_i a_i(\mathbf{x}) [\mathbf{c}_i + \mathbf{G}_i(\mathbf{x} - \boldsymbol{\mu}_i)], \quad (3)$$

where  $\mathbf{b}$  is a learned constant background. This is additive accumulation, not ordered alpha compositing. The P1 color term avoids requiring many constant-color primitives to approximate smooth gradients.

Writing  $\boldsymbol{\Sigma}_i = \mathbf{R}_i \text{diag}(\mathbf{s}_i^2) \mathbf{R}_i^\top$  makes the motion interpretation explicit. If a principal axis of  $\mathbf{R}_i$  tilts into time, the support is a sheared tube through the volume. Rendering a frame means querying the plane at its  $t$  coordinate. The representation therefore has continuous random access in space and time.

Training samples arbitrary voxels, minimizes RGB MSE, and periodically splits high-gradient primitives while pruning low-weight ones. Euclidean-center kNN supplies candidate splats; exact Mahalanobis weights are then evaluated for those candidates. Distance work is chunked to at most 100M point-center pairs, allowing 45k–120k-field experiments on an 8 GB RTX 2070 Super. Atomic checkpoints include optimizer and random-generator state, giving exact continuation across process failure and CPU/CUDA densification boundaries.

### 3.2 Gauge-free features and deterministic tokenization

Raw scale/quaternion parameters are unsuitable learning targets: quaternions double-cover rotation and covariance axes may permute. We instead use the unique log-Euclidean coordinate

$$\log \mathbf{\Sigma}_i = \mathbf{R}_i \text{diag}(2 \log \mathbf{s}_i) \mathbf{R}_i^\top. \quad (4)$$

The upper triangle contributes six values to a 22-D feature:  $[\boldsymbol{\mu}, \text{triu}(\log \mathbf{\Sigma}), \mathbf{c}, \text{vec}(\mathbf{G}), \ell]$ . Decoding by eigendecomposition reconstructs a render-equivalent field with approximately  $10^{-6}$  maximum feature round-trip error.

Dense fields are packed into a fixed  $32^3$  cell raster. The count-aware local encoder pools means, variances, counts, occupancy, and canonical rank features; the sparse decoder emits only the predicted variable number of jewels rather than allocating worst-case slots for every cell. A train-only PCA over non-overlapping  $2^3$  blocks maps eight 24-D fine cells into one 96-D coarse code. The resulting  $16^3$  grid is modeled with rotating axial attention: one block attends along  $u$ , the next  $v$ , and the next  $t$ . Three blocks communicate globally without materializing a  $4096^2$  attention matrix.

The full-field prior uses rectified flow [12]. For clean target latent  $z_1$ , noise  $z_0$ , and interpolation  $z_s = (1-s)z_0 + sz_1$ , the model regresses the velocity  $z_1 - z_0$  while receiving time and an optional CLIP condition [17]. Classifier-free dropout is required before the null branch is interpreted.

### 3.3 Persistent continuation

Long video is not generated by concatenating independent fixed clips. A stride retains jewels whose finite support intersects the overlap and copies their stable IDs and parameters exactly. The model predicts only new birth counts and 22-D birth marks in frontier-local coordinates. A  $16 \times 16 \times 8$  raster of prefix counts and moments supplies spatially aligned context tokens. This factorization turns “splat density per unit time” into an explicit birth process: the field maintains a target contributor density while old jewels persist and new regions receive births.

### 3.4 Scaffold-conditioned birth topology

The first autonomous topology head receives two aligned  $16 \times 16 \times 8$  fields. The scaffold field is trilinearly pooled RGB from the next 16 video frames. The carry field contains log density, occupancy, and mean opacity of jewels copied across the frontier. A 3D convolutional encoder with global pooled context predicts one occupancy logit  $a_j$  and one positive-count value  $q_j = 1 + \text{softplus}(r_j)$  for every cell  $j$ . Its continuous expected count is

$$\hat{n}_j = \sigma(a_j)q_j. \quad (5)$$

Training combines balanced occupied/empty binary cross-entropy, Smooth-L1 error on  $\log(1 + q_j)$  in positive cells, total log-count error, and an  $L_1$  error between normalized count distributions. The occupancy threshold is selected only on training fields by slot F1. Decoding rounds positive counts and expands each cell into nested ranks  $0, \dots, \hat{n}_j - 1$ .

Topology uses a 1,024-rank contract because one initial UCF cell contains 919 births. The frozen continuation mark flow uses 512 ranks, so coupling validates every predicted continuation cell against that smaller capacity and raises an error rather than clipping. Carried features condition the topology head but are never transformed; generated marks are appended after the exact carry.

For empty-state generation we train a second, 1,024-rank mark flow over all 72 complete UCF views: 12 initial views receive an explicitly empty context raster and 60 later views receive only the preceding causal stride. At inference, each window recomputes context and carry from the generated field, expands the topology head’s declared cell counts into ranks, samples every mark, converts frontier-local time to global time, and appends contiguous IDs. Existing rows are audited for bit-exact immutability after every append.

The observed beginning of a clip is a censored boundary, not necessarily a physical birth. A jewel assigned to initial time cell zero may therefore have finite support that began before frame zero. The projector leaves that lower boundary open only for the first local time cell of the first window; every later cell and continuation frontier remains strict. A matched control can restore the former strict initial projection without changing topology, noise, weights, or continuation semantics.

### 3.5 Stochastic mark realization and semantic guidance

The first prompt-conditioned continuation baseline predicts a sparse count raster and one canonical 22-D mark per occupied cell/rank using deterministic Smooth-L1 regression. Canonical rank provides a reproducible address, but it does not make the conditional output unimodal: several plausible future actors, poses, and foreground colors can share the same local prefix and prompt. A paired deterministic loss can therefore average mutually incompatible marks.

The controlled replacement holds target cells, counts, ranks, and carried jewels fixed and learns a rectified flow over only the marks. For data mark  $y_1$ , Gaussian noise  $y_0$ , and  $y_s = (1 - s)y_0 + sy_1$ , the network predicts  $y_1 - y_0$  while receiving the aligned prefix raster, cell/rank address, flow time, and text. A hard projection maps sampled centers and finite-support birth starts back to their declared cells without moving valid exterior edge-cell centers.

An oracle-guide variant additionally cell-pools the true future stride rendered at  $24 \times 40$ . It tests whether a coherent video scaffold contains the missing information needed to realize jewels. Because that guide contains the answer at low resolution, it measures an architectural ceiling and cannot be interpreted as text-to-video inference.

The first multiscale control constructs three guide volumes: the available future video and low-pass versions at scales (2) and (4). A  $2 \times 2 \times 2$  subgrid in every birth cell produces 24 local tokens, each containing RGB, horizontal/vertical/temporal RGB differences, local coordinates, and scale identity. A cell/rank/noisy-mark query cross-attends only to tokens in its addressed cell. For intermediate flow state  $y_s$  and predicted velocity  $v_\theta$ , the denoised endpoint

$$\hat{y}_1 = y_s + (1 - s)v_\theta(y_s, s, c) \quad (6)$$

is projected to the fixed topology and rendered on contiguous space-time patches. The tested visual objective combines RGB MSE, spatial/temporal edge L1, opponent-chroma L1, and patch SSIM while retaining feature-space flow loss for covariance stability. A follow-up hybrid keeps the entire cell-RGB 3D-convolution/global guide path and adds local tokens only as residual evidence.

### 3.6 A base-locked appearance adapter

The safe two-stream control can be compressed without granting the new module ownership of the field. Let  $v_0$  be the frozen 2.13M-parameter mark-flow velocity and let  $a_\phi$  be a compact adapter that receives the adapted mark state,  $v_0$ , addressed causal context and guide cells, cell/rank position, flow time, and text. For canonical RGB coordinates  $\mathcal{C} = \{9, 10, 11\}$ , scaffold-cell gate  $g_j \in \{0, 1\}$ , and calibrated strength  $\lambda$ , the adapted velocity is

$$v_{i,d}^{\text{app}} = \begin{cases} v_{0,i,d} + \lambda g_{j(i)} a_{\phi,i,d}, & d \in \mathcal{C}, \\ v_{0,i,d}, & d \notin \mathcal{C}. \end{cases} \quad (7)$$

The base trajectory is integrated independently from the shared Gaussian initial state. Every non-RGB value is copied from that trajectory after each Euler step, topology projection, and local-to-global transform; later topology, causal row selection, counts, density, and stable IDs also remain base-owned. The adapter has 74,067 parameters (3.48% of the base) and a zero-initialized three-channel head, so its initial sample is exactly the frozen sample. Training uses the same top 20% scaffold-saliency gate and a 288-by-192 coordinate grid that preserves the source’s 3:2 aspect ratio. A teacher-distilled arm transfers only the validated full-flow RGB velocity difference; real-video render patches remain an independent target.

### 3.7 Neighborhood-coupled jewel birth sets

The independent mark flow already shares raw noisy-mark moments through a cell raster, but after rank, noise, and guide decoding it applies every residual block row-wise. We test whether this late independence causes structured speckle by adding one permutation-equivariant set block. For hidden rows  $h_i$  assigned to birth cell  $j(i)$ , the block scatters per-cell hidden mean, variance, log-count, and occupancy; a 3D convolution mixes those statistics across adjacent space-time cells; and the addressed neighborhood state is broadcast back to each row through a residual MLP. This costs  $O(N + C)$  memory for  $N$  jewels and  $C$  cells, avoiding padded attention in initial cells that may contain hundreds of ranks.

The set block adds 414,016 parameters (+19.48%) to the 2.13M flow. Every historical tensor is loaded from the selected v1 checkpoint and frozen; only the new block trains. Its final projection is initialized to exact zero,

making the augmented velocity bit-identical to  $v_1$  before the first update. Counts, rank expansion, Gaussian paths, hard topology projection, append-only carry, stable IDs, and rendering remain unchanged. Thus the experiment attributes any difference to learned local set coordination rather than a general model fine-tune.

### 3.8 Cursor edits and constrained repair

A cursor selects jewel centers or support intersections in the parallelepiped. Translating a group creates protected moved jewels and a dirty mask covering the source, destination, swept axis-aligned bounding box, and a cell halo. Let  $m \in \{0, 1\}^K$  mark dirty latent cells and  $z^{\text{clean}}$  be the original hierarchy. After each Euler flow step, sampling reclamps

$$z_{k+1} \leftarrow (1 - m) \odot z^{\text{clean}} + m \odot [z_k + \Delta s v_\theta(z_k, s_k, c, m)]. \quad (8)$$

The final merge contains untouched jewels, generated jewels decoded from dirty cells, and protected moved jewels. This guarantees locality mechanically; visual reconciliation at the destination must still be learned from edit tuples.

## 4 Experimental Protocol

Experiments use three tiers. A synthetic moving tube falsifies the basic motion representation. CUHK Avenue [13] supplies fixed-camera fitting and generation-mechanics experiments. UCF-101 [21] supplies diverse action and appearance tests. Unless stated otherwise, the short side is 160 pixels. The current prompt corpus uses 96-frame, 120k-jewel joint fits with temporal-preserving spatial splitting.

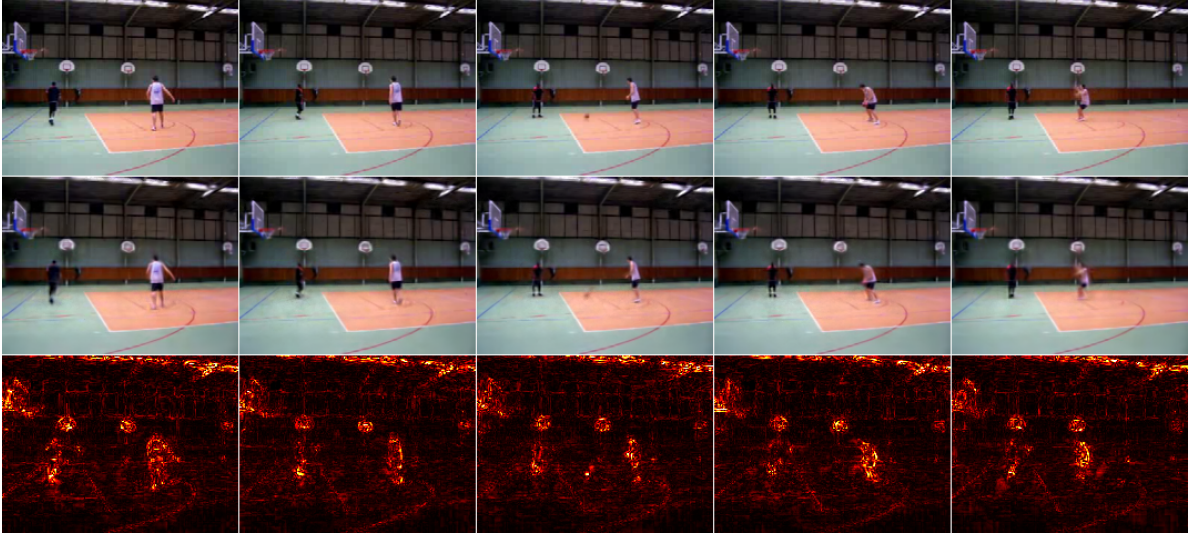
We intentionally distinguish metrics. Fit-training PSNR is often computed on sampled voxels and is not a publication-grade codec metric. “Full-volume PSNR” replays the complete saved field against the decoded source. Tokenizer PSNR compares continuous samples rendered from the fitted target and the decoded target, not decoded output against the source pixels. Energy distance evaluates latent sample distributions. Correct/shuffled/null continuation controls use identical targets and model weights. These quantities answer feasibility questions and should not be compared directly with published rate-distortion or FVD results.

## 5 Results and Discoveries

### 5.1 Reconstruction is viable, but not novel by itself

The synthetic anisotropic-tube test reaches about 55 dB. A real fixed-camera clip reaches 43.2 dB with 10k primitives; four recent UCF prompt-corpus fields reach 30.32–33.67 dB on exact checkpoint replay. Figure 2 shows the source, 120k-jewel reconstruction, and five-times-amplified absolute error for basketball. Class identity, player silhouettes, court geometry, and dominant colors survive. Error concentrates on thin limbs, the ball, faces, object edges, and motion boundaries.





**Figure 2:** Exact 96-frame checkpoint replay sampled at frames 0, 24, 48, 72, and 95. Rows are source, 120k-jewel reconstruction, and  $5\times$  mean absolute RGB error. Full replay PSNR is 33.67 dB.

The covariance is not an ornamental parameter. Median anisotropy is about 2.8–3.3 on real footage, versus about 1.2 on the synthetic background-dominated test. More importantly, attempts to impose a soft Voronoi partition reduced both reconstruction and cross-seed canonicity. On one real clip, additive splats score 43.2 dB with a 0.621 normalized cross-seed Chamfer ratio; vanilla Voronoi scores 36.8 dB/0.745. A background pseudo-cell recovers some PSNR but not canonicity; Lloyd relaxation collapses to 33.0 dB/0.784. Equalizing cell sizes fights the desired content-adaptive allocation, so the branch was removed.

## 5.2 Useful density is a rate, not a raw count

At 45k total jewels, temporal  $3\sigma$  support intersects roughly 6.2k splats per Avenue frame, but only 3.9k exceed 5% potential peak alpha and the opacity-weighted effective count is 3.3k. UCF is lower: 3.6k and 2.9k. Thus a nominally dense field remains below the desired 5k–10k effective contributors per frame.

The naive 90k isotropic control demonstrates why. It nearly doubles observed births per frame, shortens median support from 5 to 3 frames, increases effective contributors only 30.3%, and lowers PSNR. Figure 3 shows that extra count does not recover more useful detail.

**Table 2:** Matched UCF density control. The successful next control preserves temporal extent and splits spatial axes rather than shrinking every axis.

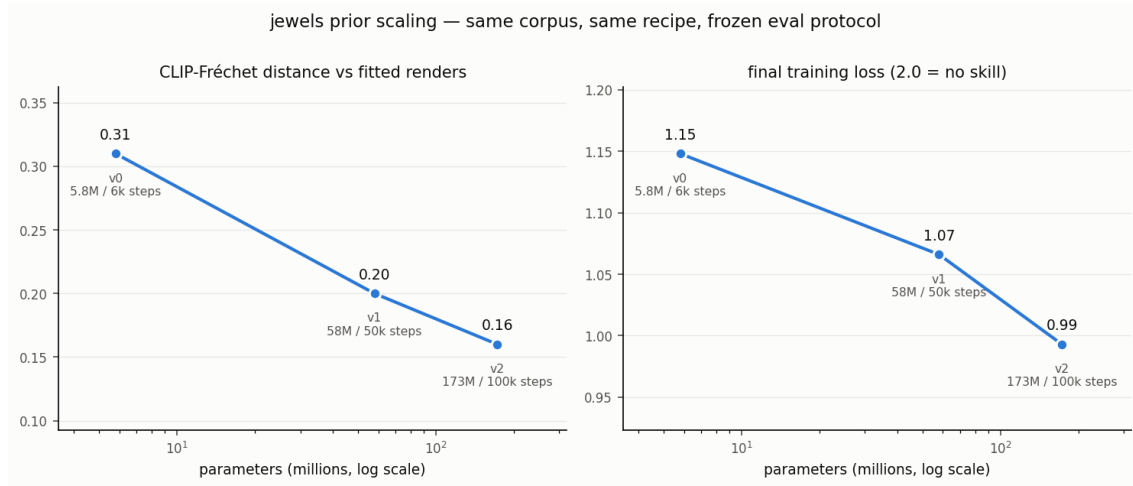
| Measurement                      | 45k isotropic | 90k isotropic | Change    |
|----------------------------------|---------------|---------------|-----------|
| Peak alpha $\geq 5\%$ /frame     | 3,595         | 4,724         | +31.4%    |
| Effective contributors/frame     | 2,933         | 3,822         | +30.3%    |
| Full-volume PSNR                 | 34.136        | 33.062        | −1.074 dB |
| Mean support lifespan (frames)   | 8.41          | 5.58          | −33.6%    |
| Median support lifespan (frames) | 5             | 3             | −40.0%    |
| Observed births/frame            | 659           | 1,290         | +95.6%    |
| Fit runtime (minutes)            | 56.8          | 79.7          | +40.3%    |



**Figure 3:** Source, 45k fit, and 90k isotropic fit at the same UCF frame. Doubling count increases short-lived fragments rather than useful detail and reduces full-volume PSNR.

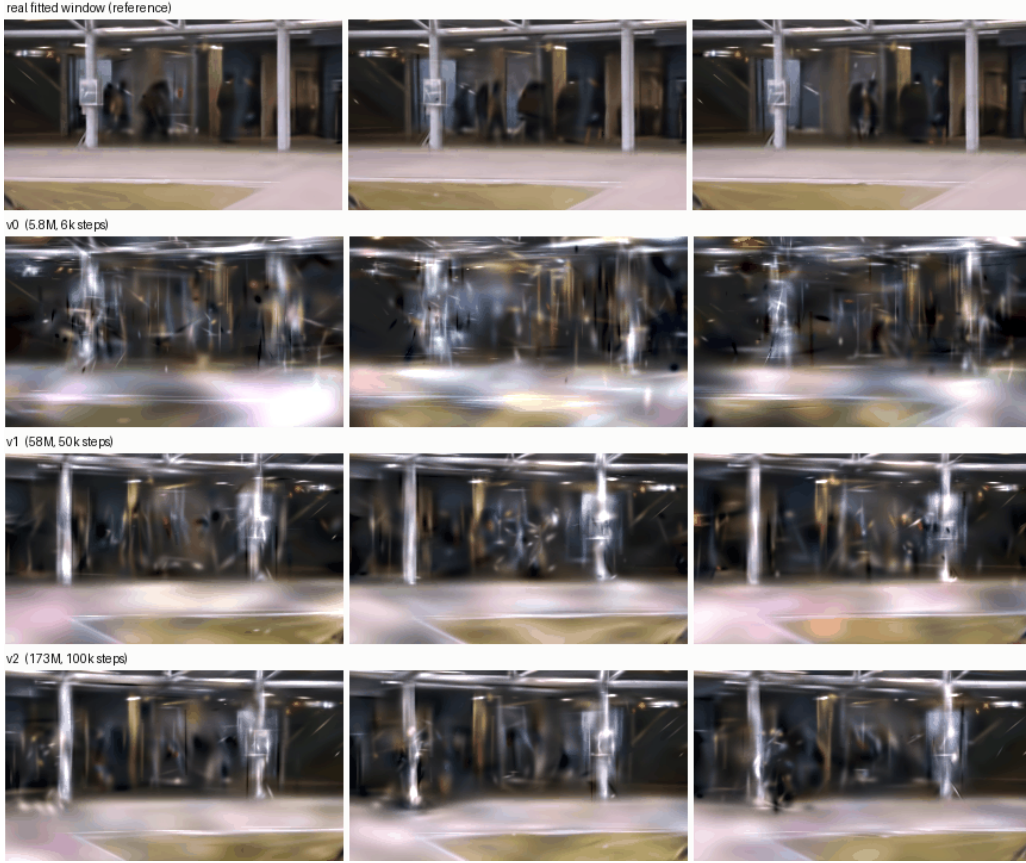
### 5.3 A generative field prior works before it works well

An early permutation-invariant SetDiT flow model directly generated 6,471-jewel Avenue fields from noise. Scaling 5.8M, 58M, and 173M-parameter models under a frozen protocol monotonically improves CLIP-Frechet distance from 0.31 to 0.20 to 0.16 and training loss from 1.148 to 1.066 to 0.993 (Figures 4 and 5). Samples preserve scene layout, palette, chirality flips induced by mirror augmentation, and temporal stability, but fine pedestrians remain distorted. The flattening curve identified data and representation density, rather than further model scaling on 231 windows, as the next axis.



**Figure 4:** Frozen scaling protocol for the first direct field prior. Improvement is monotone but saturates on the small, single-scene corpus.





**Figure 5:** A fitted reference and samples from v0, v1, and v2 at three times. Larger models preserve the coarse scene more consistently, while people remain visibly under-resolved.

#### 5.4 Locality is load-bearing for the tokenizer

On 121 dense Avenue windows, coarse cell pooling repeatedly achieved plausible aggregate PSNR while erasing people. The first source-held-out tokenizer to preserve recognizable pedestrian silhouettes uses a  $32^3$  grid, 24 values per cell, near-one-jewel average locality, and canonical-rank-aware decoding. Its 35-window audit reaches 19.987 dB and 99.14% count recovery. This remains a weak  $1.26\times$  numeric bottleneck. The  $2^3$  PCA hierarchy reduces 32,768 tokens to 4,096 and loses only 0.224 dB, making axial generation tractable.

Appearance is not solved. A bright red coat in held-out footage retains its figure but decodes tan. Rare foreground chroma is overwhelmed by uniformly weighted 22-D jewel losses and sparse uniform render samples. This is a regression-to-palette failure, not a channel-order bug. Foreground chroma and perceptual metrics must accompany aggregate RGB PSNR.

Cross-domain controls sharpen the conclusion. Frozen Avenue weights reach only 17.786 dB/86.06% count on UCF basketball; the identical architecture trained on that UCF window reaches 22.316 dB/97.13%. Equal-exposure joint training reaches 20.278 dB on held-out Avenue and 22.275 dB on UCF. The bottleneck can represent both, but absolute cell/domain specialization remains substantial.

**Table 3:** Selected tokenizer and hierarchy results. PSNR compares fitted target and tokenizer round-trip renders at fixed continuous samples.

| Representation                    | Tokens | PSNR          | Visual result       |
|-----------------------------------|--------|---------------|---------------------|
| $12 \times 12 \times 6$ , 128-D   | 864    | 18.940        | people erased       |
| $16 \times 16 \times 8$ , rank-v2 | 2,048  | 20.428        | people erased       |
| $24 \times 24 \times 16$ , pooled | 9,216  | 19.730        | faint traces        |
| $32^3$ , rank-aware, 24-D         | 32,768 | <b>19.987</b> | people recognizable |
| $16^3$ , PCA, 96-D                | 4,096  | 19.763        | people retained     |

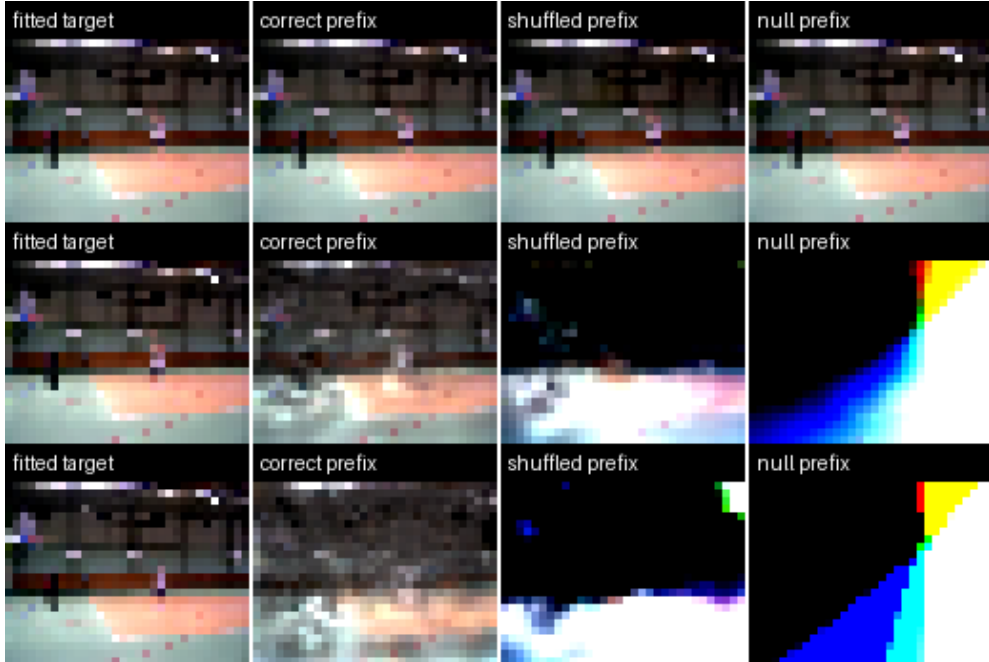
## 5.5 Generation, continuation, and editing separate into distinct gates

The 1.94M-parameter axial prior learns the coarse latent distribution: energy distance 0.2493 beats CLIP nearest-neighbor retrieval (0.3164) and repeated scene mean (0.8473). Yet correct CLIP conditioning is slightly worse than shuffled or unconditional conditioning (gain  $-0.00171$ ). With only four training cameras and one image embedding per window, the model learns scene-family statistics, not semantic control.

Continuation gives a stronger causal control. Replacing one global prefix vector with aligned  $16 \times 16 \times 8$  local tokens produces the results in Table 4 and Figure 6. Correct context reduces standardized birth-feature MSE by 62.7% against disjoint shuffled context and recovers 99.95% of births. Carried jewels are copied with zero error. This is still an overfit to four views of one clip, but it proves that spatial correspondence, not merely window identity, is required.

**Table 4:** Matched cell-local continuation control.

| Metric               | Correct       | Shuffled | Null      |
|----------------------|---------------|----------|-----------|
| Birth-feature MSE    | <b>0.4475</b> | 1.1988   | 2.1695    |
| Birth-count MAE/cell | <b>0.3310</b> | 3.7167   | 6.4994    |
| Future-field PSNR    | <b>19.870</b> | 7.302    | $-29.735$ |

**Figure 6:** Start, middle, and end diagnostics for fitted target, correct prefix, disjoint shuffled prefix, and null prefix. The extreme null output is not a video baseline because this overfit was not trained with context dropout.

The editor translates 281 real fitted jewels, dirties 448 of 4,096 coarse cells (10.94%), and merges 45,244 final jewels. Every clean coarse and fine latent is bit-identical. The repaired strip, however, is visibly washed out. A mask-aware 5,000-step branch improves dirty latent MSE only from 1.8894 to 1.8848 and does not improve the visual. The result validates the editor invariant while falsifying the current repair learner.

## 5.6 Text preflight passes; direct deterministic generation washes out

The prompt smoke test selects Basketball, HorseRiding, PlayingGuitar, and ApplyEyeMakeup from four disjoint UCF source groups: 12 training videos and four group-held-out validation videos. Three training phrasings and one unseen evaluation phrasing per class are embedded with frozen OpenCLIP ViT-B/32. Each unseen phrase retrieves the correct training-phrase centroid; the minimum correct versus best-wrong cosine margin is 0.2395 (Table 5).

**Table 5:** Held-out text geometry before jewel-model training.

| Class          | Correct cosine | Best wrong | Margin |
|----------------|----------------|------------|--------|
| Basketball     | 0.8641         | 0.5881     | 0.2760 |
| HorseRiding    | 0.9004         | 0.6610     | 0.2395 |
| PlayingGuitar  | 0.9037         | 0.6546     | 0.2490 |
| ApplyEyeMakeup | 0.9102         | 0.6229     | 0.2873 |

All 16 dense 120k-jewel fits are complete. A 0.84M-parameter direct continuation model is trained on the 12 group-1–3 videos and evaluated on all four group-4 sources. With the prefix removed, the correct unseen text template has mark MSE 1.1579 versus 1.1794 for shuffled action text and changes free decoded counts by class. Thus prompt information reaches the model, but decoded videos remain washed out and do not depict reliable action geometry.

The proposed immediate remedy was to split the sparse topology head into occupied-cell classification and positive-count regression. We instead first held the entire target topology fixed. Table 6 shows that oracle cells, counts, and ranks improve mean PSNR by only 0.172 dB. Predicted geometry and predicted color/gradient each independently collapse the render, while predicted opacity is comparatively benign. Moreover, only 39.73% of predicted centers remain in their supplied spatial cell and 25.41% in the full space-time birth cell. This is a real contract defect, but it is not the main visual cause: even the oracle-topology render remains washed out.

**Table 6:** Held-out deterministic washout decomposition averaged over four classes. “Oracle topology” supplies target cells, counts, and ranks while predicting all 22-D marks. Other rows replace only the named target mark group.

| Output                           | Mean PSNR | Finding                                        |
|----------------------------------|-----------|------------------------------------------------|
| Free deterministic prediction    | 14.060 dB | Topology and marks predicted                   |
| Predicted marks, oracle topology | 14.232 dB | Only +0.172 dB; still washed out               |
| Predicted geometry only          | 14.434 dB | Geometry error destroys structure              |
| Predicted color/gradient only    | 15.255 dB | Appearance regression independently washes out |
| Predicted opacity only           | 24.816 dB | Opacity is not the dominant failure            |

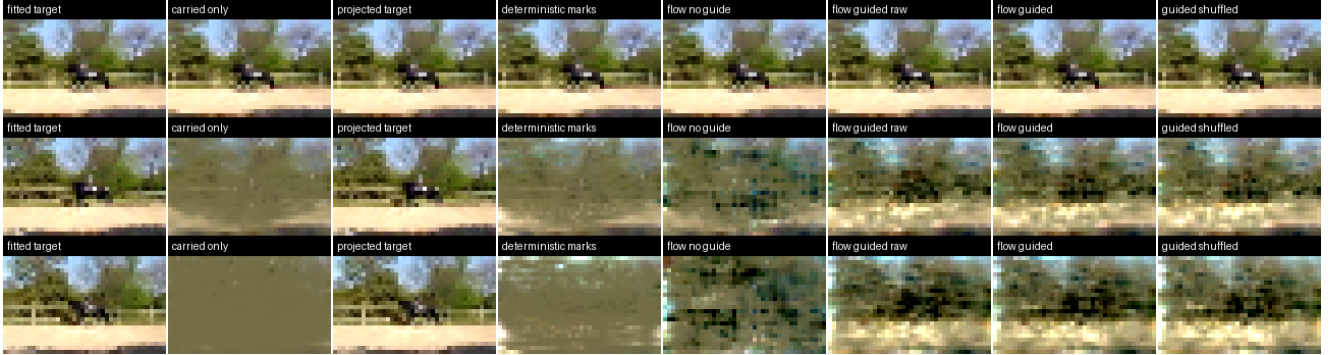
## 5.7 A stochastic jewelizer needs a coherent semantic scaffold

A 1.54M-parameter mark flow trained for 12,000 steps with oracle topology restores edge energy but not semantic coherence. Raw held-out samples average 15.064 dB and 84.4% of target edge energy, versus 14.232 dB and 59.8% for deterministic marks, yet the GIFs show structured speckle rather than recognizable action. Sampling avoids part of the conditional mean, but the local prefix and four-class text vector do not determine global scene layout.

A second 12,000-step control adds the true future only as a  $24 \times 40$  guide. Table 7 and Figure 7 show the decisive result: the guided stochastic jewelizer reaches 16.555 dB and 90.5% edge energy, recovering the court/player, horse/rider, guitar/player, and face/hand macro-layout. A corrected hard projection maps generated marks to the declared topology while leaving every fitted target render-identical at 100 dB, so the gain is not a projection artifact. Residual high-frequency noise remains substantial. Correct and shuffled text are also similar under the oracle guide because the guide already supplies the future semantics.

**Table 7:** Held-out mark realization with target birth topology. Ratios are relative to the fitted target render; the stochastic row is raw because the original projection was not target-preserving, whereas the final guided row uses the corrected projection.

| Mark generator           | Mean PSNR     | Contrast ratio | Edge ratio   |
|--------------------------|---------------|----------------|--------------|
| Deterministic regression | 14.232        | 0.574          | 0.598        |
| Unguided stochastic flow | 15.064        | 0.775          | 0.844        |
| Oracle-video-guided flow | <b>16.555</b> | <b>0.856</b>   | <b>0.905</b> |



**Figure 7:** HorseRiding at three held-out future times. Columns compare the fitted target, carried prefix alone, projection control, deterministic marks, zero-guide flow, guided raw and projected flow, and shuffled text with the same oracle guide. The guide restores scene layout that topology and stochasticity alone do not.

## 5.8 Prompt scaffolds can be fitted at the intended contributor rate

The official distilled LTX-2.3 pipeline [5] generated a balanced set of 16 prompt videos: three training phrasings and one unseen evaluation phrasing for each of the four action classes. All 16 complete at  $768 \times 512$ , 49 frames, and 24 fps. Mean generation time is 172.98 seconds and mean aggregate GPU peak is 9,399 MiB on the RTX 4090. Independent contact-sheet review preserves the requested action and primary subject in every clip, so this corpus supplies substantially more coherent prompt semantics than the direct jewel model.

Before fitting all generated videos, we scale the established 120k/96-frame budget by space-time volume, which gives approximately 69k jewels, and round to a 72k ceiling. The first Basketball evaluation-prompt fit takes 1,523.1 seconds and replays at 28.182 dB. More importantly, its opacity-weighted effective density averages 6,384 contributors/frame and ranges 5,485–7,626; 7,389/frame exceed 5% potential peak alpha. Thus it lands inside the intended 5k–10k visual regime rather than merely increasing raw total count.

The frozen contract then transfers cleanly across all four actions (Table 8). Mean full-video PSNR is 31.472 dB; mean effective density is 5,966 contributors/frame and mean 5%-alpha count is 6,883/frame. PlayingGuitar is the instructive exception: its strict effective count is 4,597/frame, slightly below the nominal floor, despite a strong reconstruction. Density is therefore retained as a content-aware diagnostic rather than enforced as a blind scalar quota. ApplyEyeMakeup reaches 35.158 dB after a checkpoint-preserving 4090-to-2070S migration and preserves face, hand, eye motion, and palette. Figure 8 shows the weakest-PSNR Basketball case: court, players, ball motion, and red uniforms survive, while fine fence/tree detail and fast limbs remain soft. These pass as semantic-scaffold targets, not as final codec results.



**Table 8:** Frozen 49-frame/72k-jewel LTX reconstruction gate. Contributor counts are per-frame means; “effective” is opacity weighted.

| Class          | PSNR (dB) | Effective | Above 5% alpha |
|----------------|-----------|-----------|----------------|
| Basketball     | 28.182    | 6,384     | 7,389          |
| HorseRiding    | 31.707    | 7,134     | 8,042          |
| PlayingGuitar  | 30.839    | 4,597     | 5,339          |
| ApplyEyeMakeup | 35.158    | 5,749     | 6,762          |
| Mean           | 31.472    | 5,966     | 6,883          |

**Figure 8:** LTX Basketball scaffold at frames 0, 12, 24, 36, and 48. Rows show decoded source, 72k-jewel reconstruction, and  $5\times$  absolute RGB error. Full replay PSNR is 28.182 dB; measured effective density is 6,384 jewels/frame.

## 5.9 Multiscale and render controls expose non-dominating trades

Table 9 reports the token-only multiscale model with render weight 2 against the cell-RGB v1 guide under identical topology, noise, sampling steps, held-out sources, and renderer. It raises target-relative contrast and edge energy, but lowers paired PSNR, mean frame-global SSIM, saturation, and temporal stability. Per-class behavior is inconsistent: ApplyEyeMakeup improves, Basketball is nearly flat, HorseRiding loses SSIM, and PlayingGuitar loses 0.948 dB. Visual inspection likewise finds cleaner local regions but no reliable action-level improvement. The branch therefore does not pass the realizer gate.

Two controls separate architecture from objective. Keeping the complete raster path and adding tokens with feature loss alone lowers every aggregate appearance metric, reaching 16.240 dB and 0.8345 SSIM. Keeping only the v1 raster architecture and applying render weight 2 raises contrast but loses 0.547 dB and 0.0087 SSIM. Reducing the predeclared weight to 0.5 improves mean SSIM by 0.0082, contrast ratio by 0.0795, and edge ratio by 0.0200, but loses 0.307 dB, reduces saturation, and increases excess temporal change. ApplyEyeMakeup improves by 0.584 dB, whereas PlayingGuitar loses 1.574 dB and overshoots both contrast and motion. The detail signal is real, but the present uniform tiny-patch estimator is not stable across actions.



**Table 9:** Matched four-class guided mark realization. Ratios target 1.0; temporal change above 1.0 indicates excess flicker/motion energy.

| Guide/objective              | PSNR          | SSIM          | Contrast      | Edge          | Saturation    | Temporal      |
|------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Raster, feature flow         | <b>16.555</b> | 0.8475        | 0.8555        | 0.9050        | <b>0.9371</b> | <b>1.7236</b> |
| Tokens, feature+render (2.0) | 16.419        | 0.8441        | 0.8694        | 0.9108        | 0.9194        | 1.7833        |
| Raster+tokens, feature flow  | 16.240        | 0.8345        | 0.8519        | 0.8925        | 0.9209        | 1.7250        |
| Raster, feature+render (2.0) | 16.009        | 0.8388        | 0.8936        | 0.9021        | 0.8836        | 1.7725        |
| Raster, feature+render (0.5) | 16.248        | <b>0.8557</b> | <b>0.9351</b> | <b>0.9251</b> | 0.9167        | 1.8055        |

The token-only model exchanged cross-cell macro context for within-cell detail, and the residual token control shows that simply stacking both paths is not sufficient at this data scale. The render controls likewise reject scalar-weight tuning as the fix: their local losses can sharpen detail while amplifying motion inconsistently. We retain v1 for the cross-domain scaffold gate and reserve future render supervision for foreground/motion-aware sampling with an explicit temporal stability term.

### 5.10 UCF-trained jewel realization transfers to unseen LTX scaffolds

Before viewing any cross-domain output, we declare that v1 must beat deterministic UCF continuation in macro PSNR and SSIM, move contrast and edge ratios toward 1.0, retain at least 98% mean birth-cell adherence, and preserve a recognizable primary subject/action in at least three of four contact sheets. Training remains the original 12 UCF sources; validation replaces only the four group-4 clips with unseen LTX evaluation-prompt scaffolds. The first 16-frame continuation view, sampling noise, target topology, and renderer are matched across controls.

Table 10 shows that the gate passes. Guided flow gains 1.565 dB and 0.2782 SSIM over deterministic continuation. Contrast rises from 0.426 to 0.743, edge energy from 0.486 to 0.830, temporal change reaches 0.981 of target, and birth-cell adherence is 99.08%. Every class improves in PSNR and SSIM: PSNR gains span 0.590 dB to 2.627 dB. Projection without the video guide is insufficient, reaching only 12.810 dB; the semantic raster is the causal input.

**Table 10:** UCF-trained mark realization on four unseen LTX scaffolds. Appearance and temporal ratios target 1.0; birth-cell values report projected adherence.

| Panel                       | PSNR          | SSIM          | Contrast      | Edge          | Temporal      | Birth cell    |
|-----------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Deterministic continuation  | 13.777        | 0.4160        | 0.4261        | 0.4860        | 0.7873        | 0.1021        |
| Flow, no guide              | 12.810        | 0.3454        | 0.5048        | 0.6691        | 0.9583        | 0.9739        |
| Flow, LTX guide             | <b>15.342</b> | 0.6942        | <b>0.7428</b> | <b>0.8300</b> | <b>0.9815</b> | <b>0.9908</b> |
| Flow, guide + shuffled text | 15.319        | <b>0.6955</b> | 0.7429        | 0.8260        | 0.9853        | 0.9904        |



**Figure 9:** One matched future time per LTX class. Columns show fitted target, deterministic UCF continuation, and guided projected flow; rows are Basketball, HorseRiding, PlayingGuitar, and ApplyEyeMakeup. Horse, guitar, and makeup retain recognizable primary subjects/actions. Basketball retains court/player layout but remains the weakest and blurriest transfer.

Correct and shuffled text are effectively tied under the true LTX guide: correct text changes PSNR by only +0.023 dB and SSIM by -0.00134. Thus the jewel model has not learned independent prompt semantics; those semantics arrive through the prompt-generated raster scaffold. This licenses the next experiment: predict occupied cells, counts, and ranks from that scaffold while stable-ID overlap state persists. Direct text-to-jewel distillation remains a later corpus-scale question.

### 5.11 Learned scaffold topology matches the oracle continuation ceiling

We train the 0.88M-parameter topology head for 3,000 updates on 72 complete initial and continuation views from the 12 UCF fields, then evaluate without adaptation on 12 views from the four LTX fits. The run takes 41.96 seconds on an RTX 2070 Super. Table 11 establishes causal scaffold use: the correct guide beats cross-class shuffled, null, and per-stride train-mean controls in slot F1 and count correlation. Removing carry drops slot F1 to 0.5374 and overpredicts births by 19.3%, making persistent overlap a load-bearing input rather than a merge detail.

**Table 11:** Held-out LTX topology over all initial and continuation views. Threshold calibration and the train-mean count field use UCF training data only.

| Topology input               | Slot F1       | Count correlation | Birth ratio |
|------------------------------|---------------|-------------------|-------------|
| Correct scaffold + carry     | <b>0.6636</b> | <b>0.6010</b>     | 0.9390      |
| Cross-class scaffold + carry | 0.6092        | 0.4728            | 0.9341      |
| Null scaffold + carry        | 0.6091        | 0.4628            | 0.8829      |
| Per-stride UCF train mean    | 0.6222        | 0.3147            | 0.9378      |
| Correct scaffold, no carry   | 0.5374        | 0.3440            | 1.1933      |

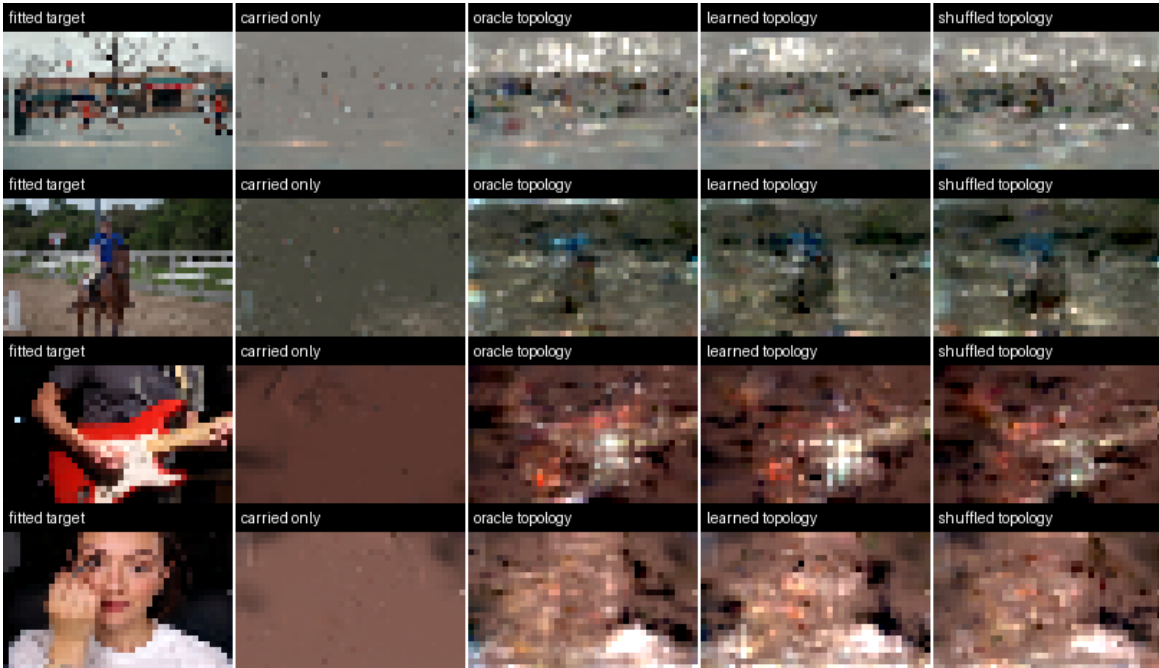
An oracle-mark rollout initially appeared to retain only 56–59% of fitted effective density: it discarded every predicted rank without an exact counterpart in one fitted decomposition. That is not a valid generative density test because additive Gaussian decompositions are non-unique. Two source-level cross-validation folds that add two LTX fits strengthen count discrimination but do not materially change exact-rank retention, arguing against fitting a larger corpus solely to fix that diagnostic.

The decisive experiment instead synthesizes every predicted rank with the frozen UCF mark flow. It evaluates the only LTX continuation view supported by the flow’s 32-frame prefix contract, frames 32–48. The first 32 frames and carried jewels are fitted seed state; the learned topology uses correct scaffold RGB and carry. For the shuffled control only the topology guide is rotated to another class; the mark flow still sees the correct LTX guide and text. Thus any learned/shuffled difference is attributable to the topology field.

Table 12 shows that learned topology predicts 70,724 births versus 70,952 targets and all four class means lie between 5,598 and 6,449 effective contributors/frame. It loses only 0.028 dB and 0.0030 SSIM to generated marks under oracle topology. Its largest cell has 113 ranks, below the frozen flow’s 512-rank capacity; spatial adherence is 100%, mean birth-cell adherence exceeds 98.8%, and carried features remain bit-identical with zero maximum error.

**Table 12:** Frozen mark realization on the held-out LTX 32–48 continuation. Appearance ratios target 1.0; density is opacity-weighted effective contributors/frame.

| Panel                                    | PSNR          | SSIM          | Contrast      | Edge          | Density      |
|------------------------------------------|---------------|---------------|---------------|---------------|--------------|
| Carried only                             | 13.529        | 0.3521        | 0.3263        | 0.3438        | 1,924        |
| Oracle topology, generated marks         | 15.492        | 0.6997        | 0.7252        | 0.8197        | 6,252        |
| <b>Learned topology, generated marks</b> | <b>15.464</b> | <b>0.6967</b> | <b>0.7202</b> | <b>0.8250</b> | <b>5,822</b> |
| Shuffled topology, correct mark guide    | 15.353        | 0.6927        | 0.7314        | 0.8170        | 5,781        |



**Figure 10:** One middle continuation frame per LTX class. Columns show fitted target, carried-only state, generated marks with oracle topology, generated marks with learned topology, and generated marks with cross-class-shuffled topology. Learned and oracle topology preserve the same macro-layout and share the same motion-region noise, localizing that residual error in mark realization.

This passes learned topology, intended density, and exact carry for one held-out continuation stride. The learned/shuffled visual margin is modest because coarse occupancy is nearly dense and the correct mark guide supplies strong semantics, although slot overlap and correlation remain causally better. It licenses the fitted-seed-free rollout below.

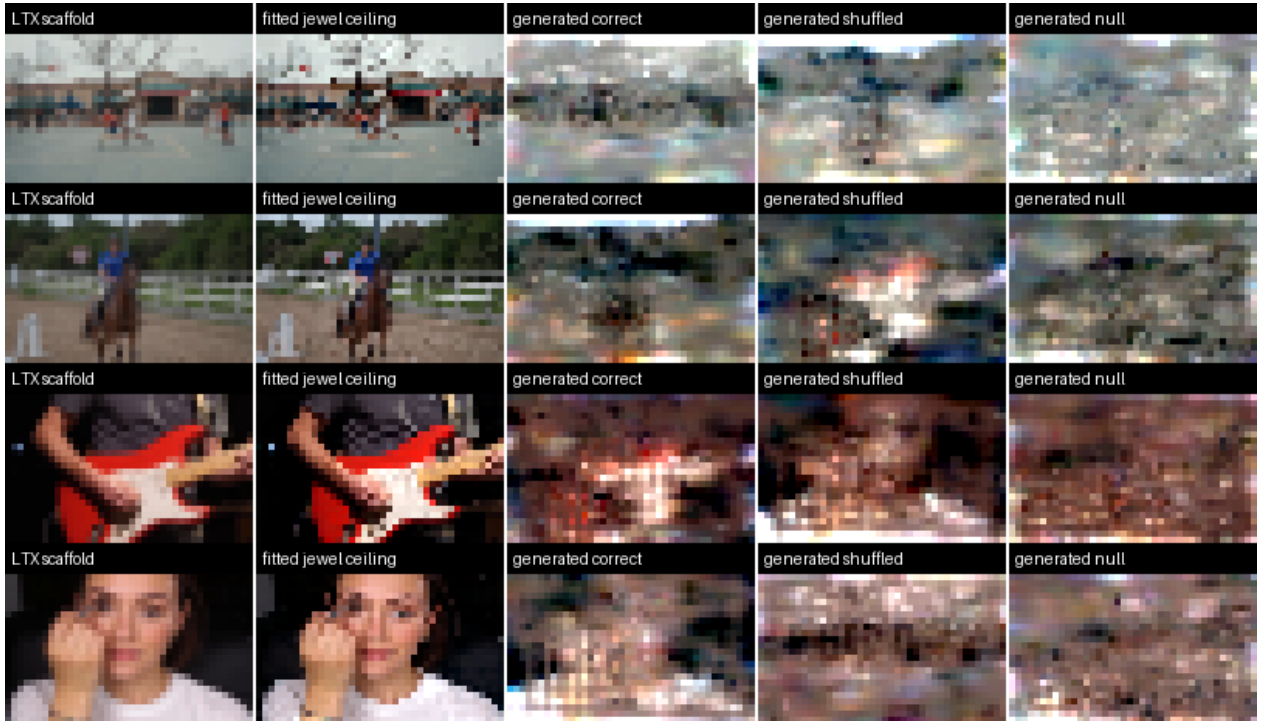
## 5.12 Empty-state generation and two free-running continuations

One 2.13M-parameter universal mark flow is trained for 12,000 updates across the 72 UCF views in 162.14 seconds on an RTX 2070 Super. We freeze it and the topology head, then generate three 16-frame windows for each held-out LTX scaffold. Correct, cross-class shuffled, and null controls share text, weights, random seed, and a background derived only from the first scaffold stride. The initial window begins from zero jewels; later context, carry, and topology state come only from the control’s own generated field. All results use deterministic PyTorch 2.12 CUDA kernels, 20 Euler steps, and seed 31.

The strict initial projection creates a mathematical opening ramp: its support-start constraint moves time-cell-zero centers to about three temporal standard deviations before frame zero, where a unit-opacity Gaussian contributes only  $\exp(-4.5) \approx 1.1\%$ . Treating frame zero as a censored observation boundary changes no learned parameter. Table 13 shows a 2.162 dB correct-scaffold gain, a larger correct-versus-shuffled margin, lower excess temporal change, and density closer to the fitted reference. Frame-zero jewels above 5% peak alpha rise from 106–244 to 8,586–9,513 across the four sources. Every prior and carried feature remains bit-identical, and every emitted ID is append-only and contiguous.

**Table 13:** Deterministic empty-state plus two-continuation rollout. Density is effective contributors/frame; seam is boundary change divided by ordinary temporal change.

| Initial boundary                | Correct PSNR  | SSIM          | Correct–shuffled | Density      | Seam         |
|---------------------------------|---------------|---------------|------------------|--------------|--------------|
| Strict finite support           | 12.408        | 0.6041        | 3.601            | 10,403       | 1.035        |
| <b>Left-censored clip start</b> | <b>14.570</b> | <b>0.6306</b> | <b>4.116</b>     | <b>7,493</b> | <b>0.964</b> |



**Figure 11:** The first fully generated field at the first frame of the third window. Rows are the four LTX classes; columns show the LTX scaffold, fitted jewel ceiling, correct-scaffold generation, cross-class scaffold, and null scaffold. Correct guidance preserves action layout, but moving subjects remain noisy relative to the fitted ceiling.

This passes the structural generation gate: all 48 frames are jewels generated from a prompt-produced video scaffold and model-produced state, effective density lies in the requested 5k–10k regime, and seams are not worse than ordinary motion. It does not pass the final fidelity gate. Correct generation remains 7.27 dB below the fitted jewel ceiling, and the actor, hands, guitar, and face contain visible motion-region noise.

### 5.13 A trivial decode of the scaffold guide bounds every pixel-fidelity claim

The rollout’s only pixel input per 16-frame stride is the  $16 \times 16 \times 8$  cell-RGB guide computed from the resized LTX video. The obvious baseline is to decode that input with no learned model at all: invert the cell raster and trilinearly upsample each stride back to render resolution. Table 14 scores this decode under the identical deterministic rollout protocol, reference, and metrics.

**Table 14:** Guide-upsample baseline versus the generated field on the four held-out LTX scaffolds (48 frames, identical protocol). Ratios are candidate/target; 1.0 matches the target’s energy.

| Arm                         | PSNR          | SSIM          | Edge ratio   | Temporal ratio | Fg. PSNR     | Quiet MAE     |
|-----------------------------|---------------|---------------|--------------|----------------|--------------|---------------|
| Guide upsample (no model)   | <b>21.081</b> | <b>0.9037</b> | 0.515        | 0.508          | <b>16.54</b> | <b>0.0064</b> |
| Generated correct (seed 31) | 14.570        | 0.6306        | <b>1.205</b> | <b>1.192</b>   | 10.23        | 0.0215        |
| Fitted jewel ceiling        | 21.840        | 0.9310        | 1.449        | 1.887          | —            | —             |

The blurred upsample wins every reference-based fidelity metric in every class, by 5.5–7.8 dB PSNR, and sits only 0.76 dB below the fitted 72k ceiling. What it cannot do is carry detail: it retains roughly half the target’s edge energy and temporal change, while the generated field restores both to approximately target level at target density. The same bound holds for the cel-shaded adaptation record (19.60 versus 15.40 dB). This is the perception–distortion tradeoff in an unusually clean form [2]: reference-based PSNR and SSIM [26] reward the blur and cannot demonstrate the generative stack’s contribution. Two consequences follow. First, any future claim of rollout quality must either beat this baseline on fidelity or be scored where detail restoration counts, i.e. with perceptual and distribution metrics such as LPIPS [29] and FVD [23]. Second, the system’s near-term value claim rests on what the baseline structurally lacks: an editable, persistent, object-decomposed field rather than pixels — and on closing the temporal-stability gap, since the trivial decode is currently  $3.3\times$  quieter in static regions than the generated field.

Scoring the same arms with per-frame AlexNet LPIPS at native  $288\times 192$  (Table 15) sharpens both consequences. The fitted 72k field reaches 0.098 LPIPS — seven times better than every other arm — so the representation’s perceptual ceiling is nearly photoreal and the entire gap lives in generative realization. The perceptual ranking also splits where the pixel ranking did not: generated fields beat the blurred baseline on the two motion-heavy classes (Basketball 0.727 versus 0.783, HorseRiding 0.703 versus 0.792) and lose on the two stable close-up classes, localizing the remaining failure to generated temporal noise on smooth content rather than missing detail. The neighborhood-coupled variant improves LPIPS over the frozen base in three of four classes, independently confirming the coupling direction. Macro LPIPS still favors the baseline by 0.015, so no perceptual victory is claimed; the fitted ceiling’s headroom identifies realization training — not the representation — as where that gap closes.

**Table 15:** Per-frame AlexNet LPIPS (lower is better) against the LTX target at native  $288\times 192$ , 48 frames, deterministic seed-31 rollouts.

| Arm                          | LPIPS         | PSNR   | SSIM   |
|------------------------------|---------------|--------|--------|
| Fitted jewel ceiling         | <b>0.0982</b> | 27.521 | 0.9831 |
| Guide upsample baseline      | 0.6761        | 19.347 | 0.8596 |
| Generated coupled birth sets | 0.6911        | 14.475 | 0.6031 |
| Generated v1 base            | 0.6971        | 14.142 | 0.6001 |

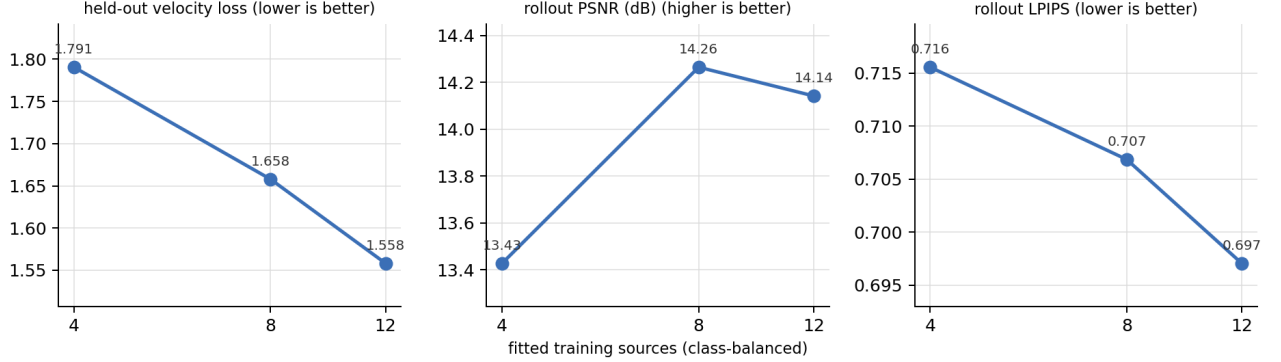
### 5.14 Fitted training data is a measured, unsaturated axis

The realization stage has never seen a data axis: the 2.13M mark flow and 0.88M topology head train in minutes on twelve fitted sources, so nothing about the current quality ceiling distinguishes “architecture limit” from “data starvation.” A three-point scaling curve separates them. Class-balanced 4- and 8-source stacks are trained with the exact 12-source recipe, seeds, and bit-identical prompt embeddings, then rolled out deterministically and scored with the identical battery (Table 16, Figure 12).



**Table 16:** Realizer data-scaling at a frozen recipe/protocol. Every arm shared across points reproduces exactly (baseline 0.6761, fitted ceiling 0.0982 LPIPS in each report), so only the training corpus varies.

| Fitted sources | Held-out velocity loss | Rollout PSNR | SSIM   | LPIPS         |
|----------------|------------------------|--------------|--------|---------------|
| 4              | 1.7906                 | 13.427       | 0.6099 | 0.7156        |
| 8              | 1.6579                 | 14.264       | 0.6041 | 0.7069        |
| 12             | 1.5582                 | 14.142       | 0.6001 | <b>0.6971</b> |



**Figure 12:** Held-out velocity loss and rollout LPIPS improve monotonically and near-linearly in log-sources with no flattening at the final point; PSNR gains 0.8 dB then stalls, the expected signature of a model gaining detail under a blur-rewarding metric (Table 14).

All three metrics improve monotonically in the mean, and a three-seed noise floor at the 12-source points (mark-flow training seed varied; topology and rollout seeds fixed) sorts the claims by significance: velocity seed-sd is 0.043 (UCF) and 0.038 (domain), LPIPS sd 0.009 and 0.013. The 4-to-8-source velocity step (0.133) is about three standard deviations and stands; the 8-to-12 step (0.050 against the multi-seed mean of 1.608) is about 1.2 and is not individually resolved; per-step LPIPS improvements are about one standard deviation each. The curve’s tail therefore needs multi-seed subset points or larger corpus steps — which is what corpus scale-up provides. The economics make the axis actionable either way: one curve point trains in under ten minutes on a consumer GPU, one fitted source costs about 25 GPU-minutes under the frozen 72k contract, and the frozen LTX teacher supplies prompt-matched source videos without limit ( $\sim 3$  minutes each). Remaining caveats: four evaluation clips, and a correct-versus-shuffled margin that thins at twelve sources.

The domain axis was then measured directly. The twelve LTX training clips, fitted under the same frozen contract (31.14 dB mean replay), replace the twelve real UCF sources with an honestly disjoint split, identical recipe, seeds, embeddings, and battery. Despite carrying half the training views (36 versus 72, since LTX fields span 49 frames rather than 96), domain-matched training cuts held-out velocity loss by a third (three seeds per arm:  $1.608 \pm 0.043$  to  $1.072 \pm 0.038$ , a thirteen-standard-deviation gap) and brings rollout LPIPS to macro parity with the guide-upsample baseline ( $0.675 \pm 0.013$  versus 0.6761; the 0.021 advantage over the UCF arm is only  $\sim 1.9$  sd and is claimed as parity, not superiority), while PSNR falls from  $14.17 \pm 0.02$  to  $12.71 \pm 0.22$ . The visual audit, however, fails: the domain-matched contact sheets lose the recognizable macro-layout the UCF-trained arm retains, dissolving into locally rich but globally unstructured texture. Patch-based LPIPS rewards the improved local statistics while being nearly blind to that layout regression, so the domain gain is established at the feature and perceptual-statistic level only, and this report treats no single metric — reference-based or perceptual — as sufficient without the rendered audit. What survives all three lenses: the cross-domain tax on the feature objective was large and is now removed, and every input to this arm’s supervision is teacher-generated — prompt-matched videos, fitted fields, and minutes of training, priced in compute alone.

The divergence itself was then diagnosed. A layout signature (average-pooling texture away at factor 8 before scoring) quantifies what the contact sheets showed: layout PSNR ranks the UCF-trained arm above mixed above domain-matched ( $15.34 \pm 0.02$  /  $14.19$  /  $13.57 \pm 0.30$  across training seeds, a six-standard-deviation structure gap) — the exact reverse of LPIPS — so the battery now reports texture and structure separately. A matched steps-per-view retrain worsens layout further, ruling out overexposure; a mixed 24-source stack interpolates between

the pure corpora on every axis, showing the two supervision sources are complementary: teacher-domain fits supply feature and texture accuracy, real-video fields supply macro-structure. Every generated arm still sits far below the trivial baseline on layout (15.7 versus 22.6 at best), making the structure crossover — not the texture crossover — the milestone the scaling program gates on per corpus doubling.

### 5.15 Foreground weighting exposes lifecycle–appearance coupling

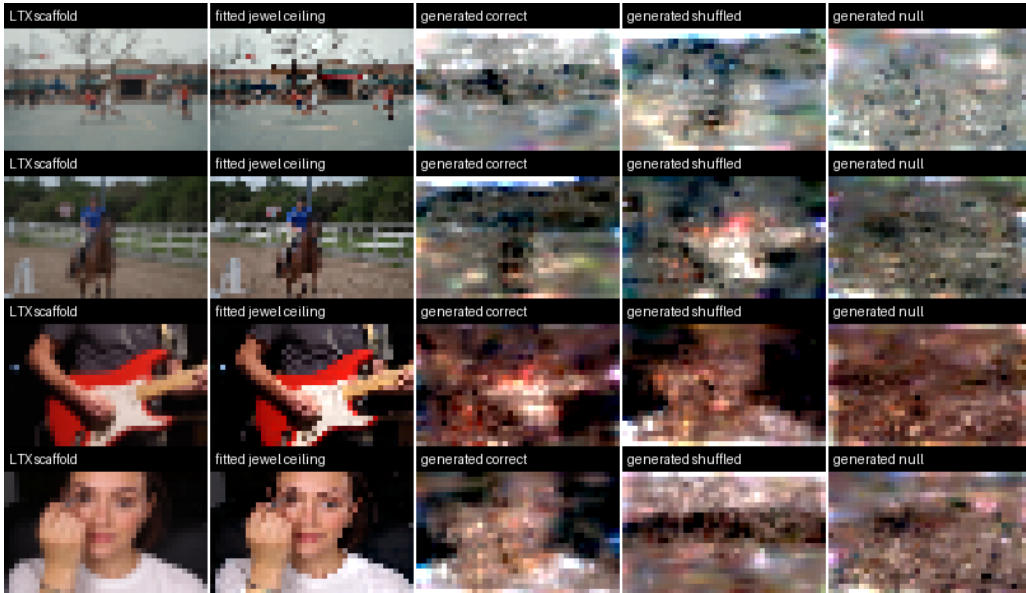
We initialize four 1,000-update arms from the passing universal flow. Target-derived scaffold saliency combines foreground/background deviation, rare chroma, spatial edge, and temporal change. Exploratory arms weight rendered salient patches, every 22-D feature, or only spatial/appearance dimensions; the final arm combines spatial/appearance weighting with a rendered quiet-change penalty. Temporal center and time-coupled covariance dimensions retain their ordinary feature loss but receive no extra saliency weight.

The decisive exact-seed comparison in Table 17 is not a null result. Global fidelity improves in all four classes; foreground-edge and motion-boundary error improve in three, and aggregate excess motion falls slightly. Yet quiet-region temporal error rises 3.8%, with PlayingGuitar and ApplyEyeMakeup each worsening by about 0.0018, while foreground PSNR falls in Basketball and PlayingGuitar. The branch therefore fails the predeclared consistency/stability gate and the base remains selected.

**Table 17:** Exact deterministic foreground/motion ablation. Lower is better for motion-boundary and quiet temporal MAE; temporal-change ratio targets 1.0.

| Arm                            | PSNR          | SSIM          | Temporal ratio | Foreground PSNR | Motion MAE     | Quiet MAE      |
|--------------------------------|---------------|---------------|----------------|-----------------|----------------|----------------|
| <b>Universal base</b>          | 14.570        | 0.6306        | 1.1918         | 10.225          | 0.07235        | <b>0.02145</b> |
| Spatial/appearance + stability | <b>14.972</b> | <b>0.6610</b> | <b>1.1872</b>  | <b>10.456</b>   | <b>0.07201</b> | 0.02227        |

The causal conclusion is architectural. Excluding lifecycle dimensions from an extra loss does not freeze them: the shared flow trunk changes all velocities, and altered appearance state feeds back into later Euler steps and continuation rasters. The next control should integrate the frozen base trajectory independently, copy its temporal center/time-covariance state exactly, and learn a separate appearance/geometry residual stream. This turns lifecycle preservation from a soft penalty into an invariant.



**Figure 13:** The rejected spatial/appearance-plus-stability arm at the first frame of the third window. Its aggregate fidelity gain is real, but the contact view does not show a uniformly cleaner subject, and the full temporal audit detects increased quiet-region flicker.

## 5.16 Exact lifecycle ownership admits a safe appearance residual

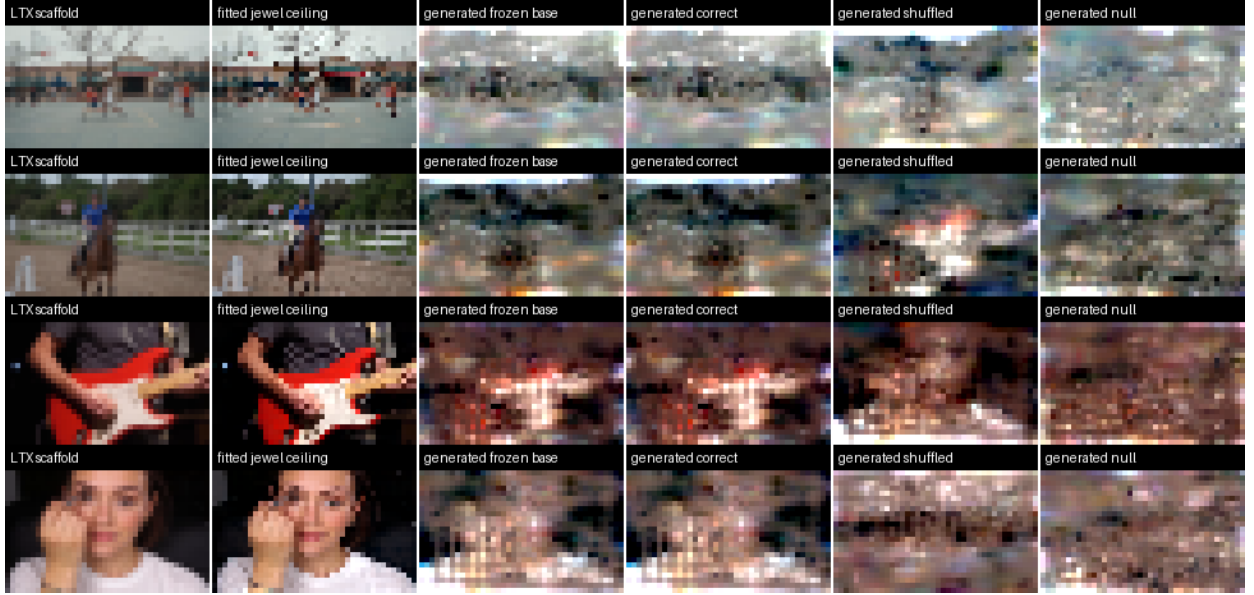
We next run the proposed two-stream control without training another checkpoint. Both streams share one Gaussian initial state, but the selected v1 flow is integrated independently and alone owns topology counts, causal row selection, stable IDs, temporal center, and time-coupled log-covariance coordinates. Candidate coordinates outside a declared residual set are copied from v1 after every Euler step, topology projection, and local-to-global covariance transform. Later topology is also predicted from v1 carry, so candidate opacity or geometry cannot change which jewels exist. The rejected saliency checkpoint supplies the second velocity field; its difference from v1 is the appearance residual. This reuse makes the experiment an architectural control rather than another optimization sweep.

Exact lifecycle copying does not by itself remove flicker. As Table 18 shows, allowing every spatial/appearance coordinate retains a 0.401 dB gain but raises quiet error 4.45%. Freezing opacity and RGB time-gradients barely changes the regression. Removing geometry cuts it to 0.32%, indicating that spatially displaced fixed-lifetime jewels still sweep through quiet pixels differently. The passing arm therefore allows only base RGB residuals in the top 20% of cells under the inference-available scaffold foreground/motion/chroma/boundary score.

**Table 18:** Deterministic lifecycle/appearance screens. The selected arm sacrifices most of the unconstrained gain to satisfy the predeclared quiet-stability invariant. Lower is better for the three MAE columns.

| Residual                      | PSNR           | SSIM           | Foreground PSNR | Foreground edge | Motion boundary | Quiet temporal  |
|-------------------------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|
| Frozen v1 base                | 14.5700        | 0.63060        | 10.2252         | 0.68952         | 0.072349        | 0.021452        |
| All spatial/appearance        | <b>14.9713</b> | <b>0.65629</b> | <b>10.6827</b>  | 0.67613         | <b>0.072201</b> | 0.022406        |
| Static spatial/detail         | 14.9148        | 0.65379        | 10.6600         | <b>0.67587</b>  | 0.072205        | 0.022388        |
| Color + spatial gradients     | 14.7476        | 0.63766        | 10.4657         | 0.68341         | 0.072240        | 0.021520        |
| Color, all cells              | 14.7550        | 0.63781        | 10.4863         | 0.68263         | 0.072224        | 0.021516        |
| <b>Color, top 20% salient</b> | 14.6041        | 0.63140        | 10.3591         | 0.68486         | 0.072268        | <b>0.021440</b> |

The selected arm gains 0.034 dB PSNR and 0.0008 SSIM, adds 0.134 dB foreground PSNR, and lowers macro foreground-edge, motion-boundary, and quiet errors without changing the 7,493 effective contributors/frame. PSNR improves in all four classes; foreground PSNR, foreground-edge MAE, and motion-boundary MAE each improve in three. Across four sources and correct/shuffled/null controls, lifecycle coordinates, count tensors, topology, and stable IDs are bit-identical in all 12 rollouts. Correct guidance remains separated from shuffled by 4.141 dB. Two class-level quiet changes remain slightly positive (below  $10^{-5}$ ), so this is a conservative factorization proof, not a claim that motion fidelity is solved. A purpose-built zero-initialized adapter must now expand the safe residual beyond RGB without relaxing exact ownership.



**Figure 14:** Selected scaffold-gated RGB residual at the first frame of the third window. The candidate is intentionally close to the frozen field: it passes stability by changing only base color in salient cells, while shuffled and null controls retain their expected semantic failure.

### 5.17 A compact adapter preserves state but only partly matches the teacher

We train the base-locked adapter on the same leakage-safe split. It has 74,067 parameters, versus 2.13M for either full flow, and trains for 3,000 updates in about four minutes on the RTX 4090. Direct target-velocity supervision improves held-out gated RGB velocity MSE by 1.53% at its best checkpoint, yet its 288-by-192 Basketball rollout loses 0.0079 dB and 0.00254 SSIM. This rejects single-time feature error as a selection metric. Distilling the validated full-flow correction instead produces a positive native-aspect result, but full strength raises Basketball quiet error by  $1.87 \times 10^{-5}$ . We therefore predeclare a one-half residual calibration for the final gate.

Table 19 reports the original 40-by-24 protocol only as an exact regression against Table 18; training and the primary visual audit use the native-aspect grid. The half-strength adapter improves its bit-identical frozen base by 0.0098 dB, raises foreground PSNR in three classes, lowers macro edge, motion, and quiet error, and keeps every class’s quiet increase below  $10^{-5}$ . Across all 12 correct/shuffled/null rollouts, every non-RGB coordinate, topology tensor, density, lifecycle value, and stable ID remains exact. Nevertheless, it captures only 29% of the full-flow PSNR gain and 33% of its foreground-PSNR gain. Full strength captures 57% of the PSNR gain but narrowly violates the class-level quiet gate and still does not match the teacher. The compact replacement criterion therefore fails.

**Table 19:** Matched four-class adapter regression. The low-resolution grid is retained solely to compare with the previously selected full-flow control; lower is better for MAE columns.

| Mechanism                    | PSNR           | SSIM           | Foreground PSNR | Foreground edge | Motion boundary | Quiet temporal  |
|------------------------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|
| Frozen v1 base               | 14.5700        | 0.63060        | 10.2252         | 0.68952         | 0.072349        | 0.021452        |
| Full-flow RGB teacher        | <b>14.6041</b> | <b>0.63140</b> | <b>10.3591</b>  | <b>0.68486</b>  | <b>0.072268</b> | <b>0.021440</b> |
| 74k adapter, $\lambda = 1$   | 14.5895        | 0.63085        | 10.3143         | 0.68647         | 0.072300        | 0.021441        |
| 74k adapter, $\lambda = 0.5$ | 14.5798        | 0.63074        | 10.2695         | 0.68799         | 0.072323        | 0.021446        |

The definitive native-aspect gate uses 288-by-192 grids and the stable per-source seeds 31–34. Relative to its independently integrated frozen base, the half-strength adapter raises macro PSNR from 14.1417 to 14.1481 dB and foreground PSNR from 9.9783 to 10.0077 dB. PSNR and foreground PSNR improve in all four classes; macro SSIM changes by  $+2.69 \times 10^{-5}$ , while foreground-edge, motion-boundary, and quiet temporal MAE fall by  $9.29 \times 10^{-5}$ ,  $1.02 \times 10^{-5}$ , and  $5.18 \times 10^{-6}$ . The largest class-level quiet increase is  $8.98 \times 10^{-6}$ , inside the predeclared limit, and all non-RGB state remains exact. Thus the adapter passes its native frozen-base structural gate. It still fails the

stronger replacement claim because the matched experiment above shows that it recovers only a minority of the selected teacher’s improvement, and the native visual difference remains minute.

The failure is informative rather than destructive. The adapter proves that architectural state ownership survives a 28.7-times smaller trainable module. What it does not prove is that isolated RGB correction solves the visible noise: the native-aspect frozen and adapted fields remain nearly indistinguishable beside the fitted ceiling. A stronger student should be distilled on complete autonomous sampling trajectories, while the main realizer must couple neighboring births and their space-time geometry rather than treating color correction as the dominant missing capacity.

### 5.18 A coupled birth-set block produces the first broad autonomous gain

We train the single zero-residual set block for 3,000 updates on the same 12-source training split, using 288-by-192 scaffold preparation and leaving the 2.13M-parameter base frozen. The run takes 24.58 seconds on the RTX 4090. Step 2,250 minimizes the fixed-path held-out error at 1.53849 versus 1.54190 for the exact base (0.221% lower) and slightly increases correct-versus-shuffled scaffold separation. The error reduction is modest, so autonomous rendering remains the selection gate.

On the exact historical 40-by-24 protocol, the coupled model improves PSNR in all four classes and raises macro PSNR from 14.5700 to 14.9751 dB. Foreground PSNR rises from 10.2252 to 11.1027 dB; foreground-edge, motion-boundary, and quiet temporal MAE fall from 0.68952, 0.072349, and 0.021452 to 0.65655, 0.071758, and 0.020750. Foreground PSNR improves materially in three classes and is flat in PlayingGuitar. SSIM falls from 0.63060 to 0.62616 and improves only in Basketball, so this aspect-sheared regression is encouraging rather than sufficient. A one-half residual calibration reduces the useful gains and still does not restore class-wide SSIM, so the full block proceeds to the predeclared native-aspect gate.

At 288-by-192, the self-consistent autonomous comparison is broadly positive (Table 20). The coupled system gains 0.334 dB macro PSNR, 0.00300 SSIM, and 0.658 dB foreground PSNR while lowering foreground-edge, motion-boundary, and quiet temporal error. PSNR improves in all four classes, SSIM in three, foreground PSNR and edge error in all four, motion-boundary error in three, and quiet error in all four. ApplyEyeMakeup loses 0.00565 SSIM; Basketball alone raises motion-boundary error, by 0.000265. Effective density remains in the intended regime at 7,797 contributors/frame versus 7,886 for the base.

**Table 20:** Native 288-by-192 birth-set screens. “Self” lets topology react to each arm’s carry; “exact count” gives the candidate the base’s complete count/rank sequence. Lower is better for the three MAEs.

| Mechanism            | PSNR           | SSIM           | FG PSNR        | FG edge        | Motion          | Quiet           |
|----------------------|----------------|----------------|----------------|----------------|-----------------|-----------------|
| Independent base     | 14.1417        | 0.60009        | 9.9783         | 0.25917        | 0.134010        | 0.023303        |
| Coupled, self        | <b>14.4753</b> | <b>0.60309</b> | 10.6366        | <b>0.25167</b> | <b>0.133451</b> | 0.022668        |
| Coupled, exact count | 14.4742        | 0.59956        | <b>10.7390</b> | 0.25175        | 0.133469        | <b>0.022582</b> |

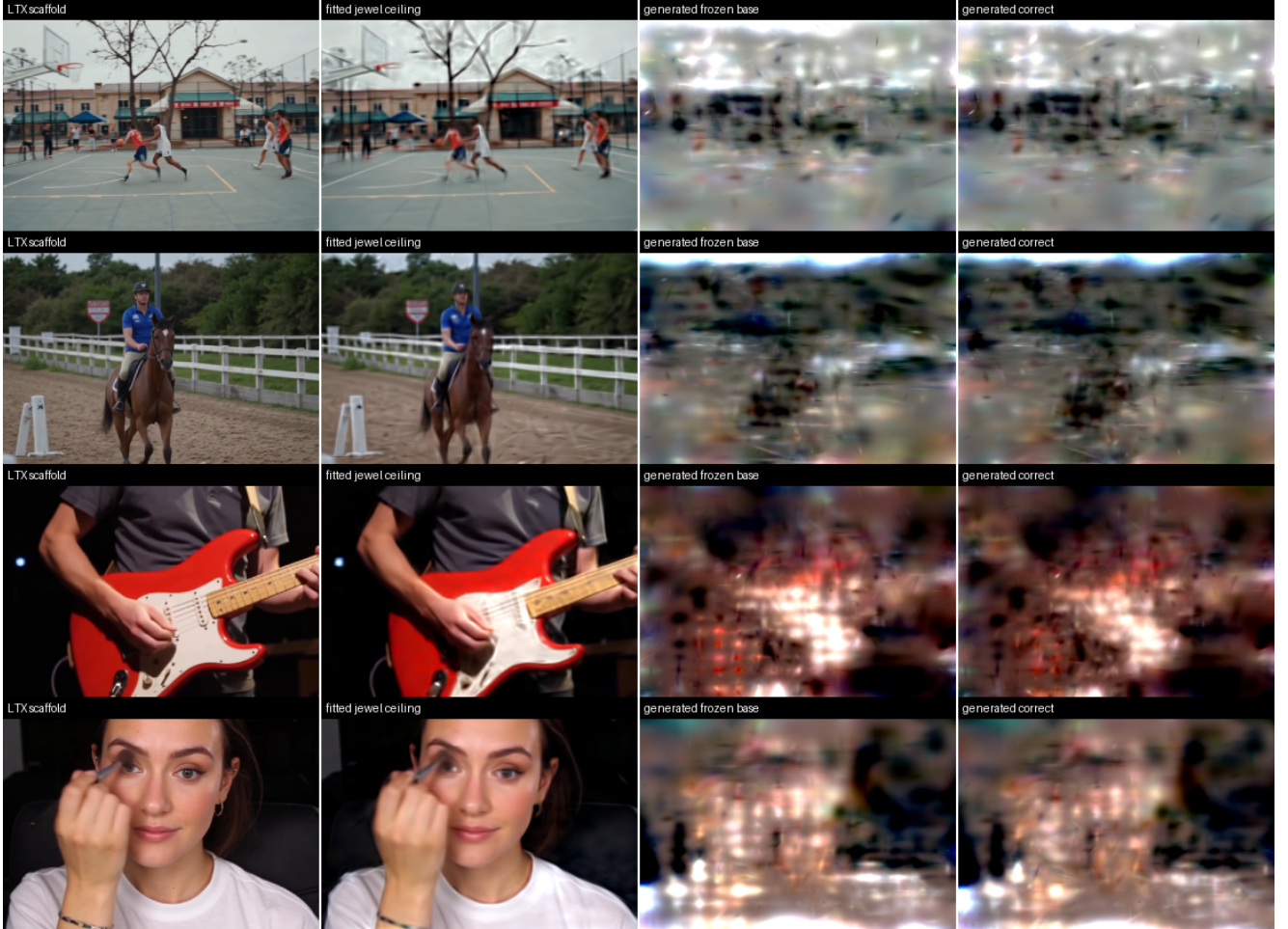
This comparison allows the frozen topology head to see each arm’s own generated carry. Later feedback therefore changes the four-class birth total by 0.70%, even though the topology model, budget policy, seeds, prompts, and renderer are shared. We consequently run a stricter attribution in which the independent base owns every per-cell count/rank sequence and the coupled candidate receives fresh identically seeded noise. All 115 inherited tensors, all 264,870 births, and stable IDs are exact between arms. This paired result retains a 0.332 dB PSNR gain and raises foreground PSNR by 0.761 dB; foreground-edge, motion-boundary, and quiet error improve in all four classes. It also exposes the limit: SSIM improves only in Basketball and HorseRiding, and ApplyEyeMakeup foreground PSNR falls by 0.035 dB. Effective density remains inside the target regime but falls 2.03% because opacity, covariance, and lifetime marks change.

Contact review is less dramatic than the metrics: the horse/rider mass and several scene boundaries improve, but the paired output is not materially more recognizable in three classes. The strict selection gate therefore fails. The experiment validates neighborhood set state as a causal direction while rejecting this feature-loss-trained checkpoint as the finished realizer; the next test should supervise rendered set/trajectory structure directly.

The convolutional mixer is not the only way to couple the set. Replacing it with an SSOG-style attention field [16] — per-atom Gaussians over relative cell displacement, content steering through bounded cold-started residuals, applied separably with occupancy riding the same three 1D passes so the field averages over born cells only — reproduces



the exact-count gains at 66k added parameters instead of 414k: foreground PSNR  $+0.775$  versus the convolution's  $+0.761$ , edge/motion/quiet all improved, PSNR  $+0.224$ , with the same family of failure (SSIM regresses; not selected). Its held-out screen gain is  $2.4\times$  the convolution's on a matched same-device battery, and the learned geometry is directly readable: three of four atoms look backward in time with temporal reach wider than spatial — self-discovered temporal-context gathering — and all three cold-started steering gates open during training, the width gate hardest. One methods caveat these comparisons surfaced: seeded evaluation noise paths differ across GPU architectures, offsetting absolute battery values by more than typical model deltas, so all screens here are same-device.



**Figure 15:** Exact-count native audit at the third-window frontier. The coupled field (right) shares every birth cell/rank and noise seed with the frozen base (center-right). Changes are real but subtle relative to the large gap to the fitted-field ceiling.

### 5.19 A feed-forward encoder replaces mark-space emission

The accumulated evidence — layout below the trivial-decode baseline for every sampler, a data curve that moved texture but never structure, and coupling gains an order of magnitude too small — identifies joint composition under additive rendering as the failure mode of direct set generation, and a visual review of the rollouts made the verdict operational: the mark-space path is retired as primary. The replacement inverts the architecture. A  $\sim 5\text{M}$  feed-forward encoder performs *amortized fitting*: slots are seeded from the video itself on a stratified per-cell lattice with widths and opacities calibrated so the initial field renders the video softly at unity coverage, and a 3D-convolutional trunk predicts bounded refinements, trained with the fitter's own stochastic-voxel render loss on teacher-generated (video, fitted-field) pairs — a supervised regression with manufacturable data. Seeding is load-bearing: the identical model without it underfits at 12.2 dB.

**Table 21:** One-shot encoding of the four held-out clips at native  $288 \times 192$  (final step-8000 checkpoint; the mid-run step-3000 snapshot already scored 23.24 / 0.4693). The encoder is the first arm to beat the trivial decode on every metric.

| Arm                                           | PSNR         | SSIM          | LPIPS         | Layout PSNR  |
|-----------------------------------------------|--------------|---------------|---------------|--------------|
| Fitted jewel ceiling (9,000 opt. steps/clip)  | 27.52        | 0.9831        | 0.0982        | 30.46        |
| <b>Encoded one-shot (single forward pass)</b> | <b>23.41</b> | <b>0.9436</b> | <b>0.4049</b> | <b>30.14</b> |
| Guide upsample baseline                       | 19.35        | 0.8596        | 0.6762        | 22.61        |
| Best mark-space generative arm                | 14.48        | 0.6031        | 0.6911        | 15.75        |

Macro-layout reaches the fitted ceiling — the deficit that defined every generative arm is eliminated — and the remaining 4.3 dB is uniform softness with three supervised levers: corpus (twelve windows here; the teacher mints more at  $\sim 30$  GPU-minutes each), capacity, and refinement passes. The system prompt  $\rightarrow$  teacher video  $\rightarrow$  encoder  $\rightarrow$  persistent jewel field is thereby end-to-end real: semantics at teacher quality, persistence, identity, and editability at the field level. Set-space emission continues as a research lane in latent-set form, with the steered Gaussian coupling of the previous subsection as its substrate.

## 6 What Has Been Falsified

Failures have been used as architectural evidence rather than hidden tuning history:

**Table 22:** Selected falsifications and the resulting decision.

| Failed hypothesis                                              | Decision                                                                                                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Soft Voronoi improves canonicity                               | Removed; additive wins reconstruction and normalized cross-seed Chamfer.                                                            |
| More raw splats automatically increase useful density          | Rejected; audit contributor rate and preserve temporal axes during split.                                                           |
| Aggregate PSNR selects a visually useful tokenizer             | Rejected; require held-out motion silhouettes and foreground chroma checks.                                                         |
| Global prefix context is sufficient for births                 | Rejected; aligned cell-local context improves future-field PSNR by 3.41 dB.                                                         |
| The six-camera CLIP prior uses semantics                       | Rejected; correct condition does not beat shuffled/null.                                                                            |
| Sparse count/topology error is the primary washout cause       | Rejected; exact target topology adds only 0.172 dB.                                                                                 |
| Stochastic marks alone recover coherent video                  | Rejected; edge energy returns, but samples remain structured noise without a video guide.                                           |
| Local multiscale tokens automatically improve the raster guide | Rejected; token-only and residual-hybrid controls both lose the matched visual gate.                                                |
| Scalar render-loss tuning cures washout                        | Rejected; detail improves at weight 0.5, but one action becomes less faithful and less stable.                                      |
| Scalar foreground weighting separates detail from lifecycle    | Rejected; global fidelity rises, but quiet-region temporal error rises 3.8%.                                                        |
| Exact lifecycle copying alone removes quiet-region flicker     | Rejected; full-field appearance still changes rendered motion, so the passing residual is scaffold-gated RGB only.                  |
| Single-time RGB distillation fully replaces the second flow    | Rejected; the 74k adapter preserves exact state but captures only 29% of the teacher’s gated PSNR gain at the passing strength.     |
| Independent post-guide jewel rows are sufficient               | Rejected; one frozen-base neighborhood set block improves exact-count native PSNR and all three error measures in all four classes. |
| Exact fitted cell/rank overlap measures generated density      | Rejected; synthesizing all predicted ranks restores 5,822 effective contributors/frame and oracle-level rendering.                  |
| Mask visibility alone teaches repair                           | Rejected; train explicit edit tuples with protected-jewel summaries.                                                                |
| Further model scaling solves the sparse prior                  | Deprioritized; the curve saturates on 231 single-scene windows.                                                                     |

## 7 Limitations

- **No direct text-selective jewel generator.** The hybrid path now generates all 48 jewel frames from a prompt-produced LTX scaffold and its own state, but the jewel model’s correct and shuffled text controls remain tied because semantics live in the raster scaffold.
- **Small and specialized corpora.** The generative results use one fixed-camera scene; the continuation result is an overfit; the prompt corpus contains only 16 videos.
- **Fixed camera and no cuts.** Global camera motion shears most primitives at once and spends density on camera motion. Camera tokens or explicit stabilization are future work.
- **Low resolution.** Current fits use a 160-pixel short side. PSNR values are not comparable to full-resolution codec papers without a matched rate-distortion protocol.
- **Conservative lifecycle factorization.** Exact two-stream ownership passes the macro stability gate only when the residual is restricted to RGB in the top 20% scaffold-salient cells. The dedicated 74k adapter preserves that ownership but only partly matches the full-flow correction; complete-trajectory distillation and broader residuals remain open.
- **Short autonomous horizon.** Fitted cells, ranks, marks, and carry are removed for all three generated windows, but the audit covers only 48 frames, four action classes, and one seed. Longer rollouts may amplify carry error

or exhaust the fixed topology/rank contract.

- **Weak rendered topology selectivity.** Correct scaffold topology beats shuffled/null in count space, but shuffled-topology rendering is only 0.111 dB below correct because occupancy is nearly dense and the mark flow still sees the correct guide. Finer/adaptive density fields may be needed for topology itself to control semantics.
- **Motion-region mark noise.** Learned and oracle topology share blur/noise on moving actors, hands, and thin objects. The coupled set block produces a broader 0.334 dB native gain, but reaches only 14.475 dB against a 27.521 dB fitted-field ceiling and remains visibly speckled. Cell mean/variance coupling is therefore useful but insufficient.
- **Pure-PyTorch rendering.** Exact kNN/Mahalanobis evaluation is useful for research but slower than a tile CUDA rasterizer. Current reconstruction is optimization-based, not amortized.
- **Edit quality.** Clean-region identity is exact, but dirty-region synthesis is not yet visually useful and may create additive collisions with protected moved jewels.
- **Reference-based fidelity metrics saturate below a trivial baseline.** A no-model trilinear decode of the scaffold guide beats every generated rollout on PSNR, SSIM, foreground PSNR, and quiet-region stability (Table 14); the generated field wins only detail- and motion-energy restoration. Until perceptual/distribution metrics (LPIPS, FVD) and external baselines are in place, no fidelity number in this report should be read as evidence that the generative stack outperforms trivial decoding of its own input.
- **Novelty is provisional.** Closely related literature is moving quickly. A formal claim requires a complete system, broader search, matched baselines, and external review.

## 8 Why an Explicit Jewel Field Rather Than a Raster Latent?

The relevant comparison is not “jewels versus diffusion.” A jewel field specifies the state that is represented and rendered; diffusion, flow matching, and autoregression specify how a model learns or samples a distribution over states. Video diffusion models usually operate on dense pixel or learned spatiotemporal latent grids [1, 6]. The rectified-flow prior used here is itself closely related: flow matching supports Gaussian probability paths that include diffusion paths as special cases [12]. A future JEWELS model could therefore use a diffusion or flow objective while retaining an explicit field as its decoded state.

Dense raster latents have formidable advantages: regular tensors map efficiently to accelerators, convolutions and attention are mature, topology changes are implicit, and large pretrained image and video priors already exist. The case for jewels is instead a set of interaction and state properties summarized in Table 23. These properties do not by themselves imply better photorealism or rate-distortion.

The oracle-guide experiment argues for using both representations where each is strongest. A pretrained raster video model can supply global scene semantics, pose, camera motion, and prompt knowledge; a stochastic jeweler can convert that coarse scaffold into explicit persistent state. This is not a concession to pixel-only output: users still select, move, carry, render, and repair the final jewels. It is a bootstrap strategy that avoids asking a 12-video point-set learner to rediscover a video foundation model.

**Table 23:** Representation-level hypotheses relative to an opaque raster video latent.

| Field property                | Potential capability                                           | Required evidence                                               |
|-------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------|
| Stable primitive identity     | Exact carry across generation windows                          | Held-out autonomous rollouts without identity drift             |
| Localized finite support      | Hard-clamped local repair after direct manipulation            | Perceptually convincing edits with unchanged clean regions      |
| Adaptive primitive density    | Spend capacity on motion, boundaries, and rare detail          | Matched quality and rate at 5k–10k effective contributors/frame |
| Continuous (u,v,t) domain     | Random-access time slices and resolution-independent rendering | Quality and speed against matched video decoders                |
| Explicit marks and covariance | Select, move, recolor, split, or delete native state           | User edits that do not require global video regeneration        |

The present motion noise is therefore a diagnostic, not something raw scaling is assumed to erase. The 45k-to-90k isotropic control increased effective contributors by only 30.3%, shortened support lifetimes, nearly doubled births,

and reduced PSNR. More jewels should sharpen silhouettes only when densification splits spatial axes while preserving temporal support, and when motion-, edge-, and foreground-weighted samples prevent static background from dominating the objective. More fitting steps can let new primitives settle, while a shared generative prior may suppress inconsistent fit noise; neither can restore detail discarded by the representation or tokenizer. The direction should be rejected if matched temporal-preserving density fails to reduce motion-boundary error, if the tokenizer approaches one token per jewel, or if local edits still require near-global repair.

## 9 Jewel Fields as a Multimodal Event Language

Native multimodal generation is established prior art. Chameleon models arbitrary interleaved image and text token sequences [3]; Emu3 applies next-token prediction to discrete image, text, and video tokens [25]; and Transfusion combines language modeling and continuous image diffusion in one transformer [30]. We therefore do not claim that an LLM generating visual media is new. The narrower opportunity is to make the generated visual tokens persistent, spatially executable state rather than only codes that decode to a raster.

A jewel video admits an event factorization such as

$$\text{WINDOW}, \text{CARRY}(i), \text{CELL}(c), \text{BIRTH}(n), \text{MARK}(\theta), \text{MOVE}, \text{SPLIT}, \text{DELETE}, \quad (9)$$

where  $i$  is a stable primitive identity,  $c$  is a space-time cell,  $n$  is a local birth count, and  $\theta$  contains continuous position, covariance, opacity, and color marks. Text, audio events, camera instructions, selections, and edit commands could be interleaved with this stream. The model output would then be both media and an executable edit program: retained state is copied exactly, new events extend the frontier, and an edit dirties only the affected cells.

A flat sequence of 120k jewels per clip would be impractical and would discard spatial locality. The selected hierarchy suggests a tractable factorization: autoregress or mask-model the  $16^3$  coarse cell tokens and discrete event/count symbols, then use a local continuous head or conditional flow to emit fine 22-D jewel marks inside occupied cells. This also avoids pretending that every jewel is already an object-level semantic token; current primitives are low-level basis functions, and stable IDs are not automatically object identities. The multimodal-language hypothesis succeeds only if the hierarchy achieves substantial token compression, text controls held-out motion rather than retrieving a training clip, persistent events survive long rollout, and the same structured state supports cheaper local editing than full latent regeneration.

## 10 Path to a Promptable Model

The shortest evidence-driven path is:

1. freeze the passing two-stream control: v1 owns topology/lifecycle/IDs exactly, while RGB residuals are confined to the top 20% scaffold-salient cells;
2. retain the 74k zero-initialized adapter as a structural proof, but do not call it a full-flow replacement: at its passing strength it reproduces only part of the teacher correction;
3. if compact residual compression remains useful, distill complete autonomous teacher trajectories before expanding beyond RGB, and retain the class-level quiet gate;
4. address the larger visual bottleneck with a cell- or neighborhood-coupled set realizer whose births share local space-time geometry rather than sampling every addressed rank independently;
5. extend the autonomous audit beyond three windows and across multiple prompts/seeds before interpreting exact short-horizon carry as long-video stability;
6. evaluate finer or adaptive topology only if correct-versus-shuffled rendering remains tied after the mark guide is controlled;
7. train edit tuples with moved protected jewels, dirty masks, and local scaffold crops so source/destination repair is learned while clean regions remain exact;
8. build a 1k–10k captioned jewel corpus and distill the scaffold pathway toward direct text-to-jewel generation after the coupled realizer passes prompt and rollout gates.

A publication-grade comparison should include GaRV/VeGaS/GSVC-family reconstruction controls, pixel-space video baselines, parameter count, fitting/encoding time, memory, random-access render speed, rate-distortion, FVD or VBench-style generation metrics, edit locality, and prompt selectivity. The unique value proposition



is not winning every codec metric; it is combining continuous temporal structure, persistent generation, direct manipulation, and constrained repair in one native representation. A longer-term foundation-model path would expose the coarse space-time hierarchy and event vocabulary to a multimodal transformer while retaining a continuous local generator for fine jewel marks.

## 11 Conclusion

The broad idea that Gaussian splats can reconstruct video is established prior art. The present work contributes a disciplined attempt to turn that representation into a generative and editable medium. The strongest findings are structural: anisotropic space-time support is genuinely used; effective density depends on temporal lifetime rather than raw count; near-one-jewel tokenizer locality preserves small actors; spatially aligned prefix context is essential for births; and local editing can preserve untouched latents exactly. The new washout audit adds a crucial causal result: correcting topology does not correct appearance, stochastic marks restore detail without semantics, and even a tiny coherent video scaffold substantially restores layout. The cross-domain gate strengthens that last result: without retraining, the UCF jeweler improves all four unseen LTX classes over deterministic continuation and preserves three recognizable actions.

The project is therefore past the representation-mechanics question but before the text-to-video claim. It is also past the one-stride target-topology question: learned scaffold/carry counts feed the frozen mark flow at the intended density and reach the oracle-topology ceiling. It is now also past the fitted-seed-free short-rollout question: an empty initial field plus two continuations preserve append-only identity at the intended density, and a matched boundary control explains the former opening washout without retraining. The next milestone is not another scalar loss sweep. Foreground/motion weighting yields real detail but trades it against quiet-region stability, so the two-stream control separates an exact frozen lifecycle trajectory from appearance. It passes only under a conservative scaffold-gated RGB residual: topology, density, lifecycle, and stable IDs stay exact, three-class subject metrics improve, and macro quiet error falls, but the gain is just 0.034 dB. The zero-initialized compact adapter now preserves that ownership with 3.48% as many parameters as the base, but at its passing strength recovers only 0.010 dB and 29% of the teacher’s gain. A first neighborhood-coupled set block now validates the proposed direction: with the base frozen and exact base-owned counts it gains 0.332 dB at native aspect, improves foreground PSNR by 0.761 dB, and lowers all three macro error measures. Yet it misses the three-class SSIM criterion, the contact-sheet change remains subtle, and the fitted-field gap remains 13.05 dB. The next realizer step should therefore train richer set geometry or trajectory-level rendered structure, not merely extend the same row-wise feature loss; complete-trajectory adapter distillation is a secondary compression problem. The hybrid preserves the distinctive explicit state while inheriting semantic competence that the current corpus cannot teach; prompt semantics still live in the pretrained scaffold, and direct distillation remains a later gate.

## A Reproducibility Snapshot

**Table 24:** Current reproducibility-critical constants.

| Component          | Frozen or selected setting                                                                     |
|--------------------|------------------------------------------------------------------------------------------------|
| Coordinate domain  | normalized $[-1, 1]^3$ in horizontal, vertical, and time                                       |
| Primitive feature  | 22-D: center, log-covariance, P0 RGB, P1 color gradient, weight logit                          |
| Renderer           | additive P1, learned background, kNN 64, 100M distance-pair cap                                |
| Dense prompt fit   | 96 frames, 160-pixel short side, 16k init, 120k maximum, spatial split                         |
| Fine tokenizer     | $32^3$ cells, 24-D local latent, sparse variable-count decode                                  |
| Hierarchy          | non-overlapping $2^3$ train-only PCA, 96 retained components, $16^3$ codes                     |
| Prior              | rectified flow, rotating axial attention, nullable CLIP/text condition                         |
| Continuation       | empty initial state plus two 16-frame strides; generated-state context/carry; append-only IDs  |
| Topology generator | 0.88M 3D cell-RGB/carry head; $16^3$ cells, 1,024-rank initial contract                        |
| Mark generator     | selected 2.13M 1,024-rank 22-D rectified flow; 20 deterministic Euler steps                    |
| Appearance control | independent frozen v1 trajectory; RGB residual in top 20% scaffold-salient cells               |
| Compact adapter    | 74,067 parameters; RGB only; $\lambda = 0.5$ passing quiet calibration; exact base ownership   |
| Semantic control   | prompt-generated LTX $24 \times 40$ guide; three-window autonomous rollout passes structurally |
| Editing            | swept source/destination dirty mask, protected moved jewels, exact clean-cell clamp            |

## B Novelty Checklist Before Submission

1. Update the literature search through the submission date, including citations that cite VeGaS, GaRV, HyperGS, and Diff4Splat.
2. Replace the provisional positioning paragraph with a feature-by-feature comparison verified against code or supplements, not abstracts alone.
3. Demonstrate text-conditioned generation from noise on held-out prompts and sources.
4. Extend the passing initial-plus-two-continuation result across longer horizons and multiple seeds/prompts; replace one-step adapter distillation with trajectory supervision or document the compact branch as a structural control only.
5. Demonstrate a moved-object repair in which clean regions are exact and the dirty region is perceptually plausible under multiple samples.
6. Release the protocol, checkpoints, videos, and a matched baseline script.

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